Matheus Janczkowski

Introduction to continuum mechanics

an incursion into computational mechanics self-study

Florianópolis, Brasil
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A didactic introduction to the basic concepts of continuum mechanics and their mathematical foundations for self-study.

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This work is intended for educational, academic, and scientific use.

Preface

This work is part of a series on self-study. The series of self-study aims to teach programming, mathematical analysis, continuum mechanics, and numerical methods for boundary value problems to a wide range of students, from an undergraduate background to the graduate student, as a supplementary material. The main objective is to allow the reader to teach him or herself with little to no external help. This objective is framed by the use of analogies and a conversational prose, as well as a stronger basis on mathematics.

We believe that undergraduate courses on mechanical engineering lack disciplines on continuum mechanics, which, in turn, hinders the analytical ability of students to tackle real-world problems. Numerical methods, such as the finite element method are widely available through commercial softwares, such as *Ansys* and *Nastran*. Nonetheless, the understanding of the users is too little to critically analyze numerical results. This undesirable reality might even pose risk to the practice of engineering and design activities.

Given this context, we aim to introduce the reader to the underlying principles that govern physics and numerical analysis. We attempt to present each and every concept as amenable as possible, without losing physical and mathematical precision. The next text of the series will be dedicated to introduce the finite element method as means to approximate solutions to boundary value problems stemming from the physical principles here defined.

Mathematical derivations were reordered (when compared to classical literature) to highlight the style of this series—mathematics are fundamental tools to understanding the underlying physics, hence it must be seen alongside physics; not apart from it. Exercises and challenges are proposed throughout and sprinkled around the text. Therefore, the reader might test his or her skills as he or she progresses over the text.

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Matheus Janczkowski

Contents

Part I

Continuum mechanics

Continuum mechanics is a mathematical and physical framework for modeling and describing the motion and interactions of matter within classical mechanics. Classical mechanics is usually presented in the scope of point masses or rigid bodies. Continuum mechanics, in contrast, extends the definitions and techniques of classical mechanics to deformable bodies—solids, liquids, and gases. Continuum mechanics encompasses different physical phenomena as well, such as electromagnetism, incompressible flow, and heat transfer. The most fundamental hypothesis is the ability to divide matter into infinitesimal chunks around an arbitrary point. The consequence is that the domain is continuous and calculus can be used to formulate physics into mathematical equations.

This part of the text is dedicated to introducing the mathematical framework required to model nature. We will define the fundamental concept of tensors and then we will perform calculus with them. Motion, deformation, and strain of a generic body will be defined, as well as external forces acting on it. At the last two sections, conservation laws and the second law of thermodynamics will be presented. Finally, one will be able to derive boundary value problems upon deformable bodies.

1. Algebra with tensors

1.1. What are tensors?

Tensors are mathematical entities that, broadly speaking, have their physical meaning unchanged by different choices of coordinate systems. Examples of this concept are pressure, displacement, stress, and etc. Even if the observer changes its position in space, the physical significance of each one of those quantities remains the same.

Following this intuition, one should be worried, since pressure, displacement, and stress are completely different things; they even need a different **quantity of numbers** to represent each one of them. Hence, that is precisely the reason to why a tensor has **order**—a scalar, such as pressure, is said to be a *zero-order* tensor; a vector, exemplified by displacement, is a *first-order* tensor; stress and strain are examples of *second-order* tensors. Order, in summary, tells the number of **indices** necessary to portray each component with respect to directions in space (more on that later).

The concept of order hints us that tensors can be also seen as **arrays of numbers**. As we will duly see in the next sections, each one of those number are called the **components** of a tensor. A simple example at hand is velocity $\boldsymbol{v}$: in a three-dimensional space, with coordinates (x, y, z), there are velocity components in the three canonical directions—v_x, v_y, and v_z. The definition given at the beginning is closely tied to the concept of components. If the system of coordinates changes, the components of a tensor change according to an appropriate transformation, whereas the tensor remains unmodified.

As arrays of numbers, tensors are particularly useful for evaluating derivatives. An illustrative example is the derivative of a vector field with respect to the position vector in space, for each component of the vector field must be differentiated with respect to each component of the position vector (i.e. with respect to each space coordinate). That example operation is the gradient, to which the reader will be quite acquainted in the near future. In summary, to evaluate the gradient of a tensor, such as vector, all combinations of derivatives of the tensor components with respect to the spatial coordinates must be evaluated. If higher-order tensors are to be differentiated, the concept of tensors becomes unavoidable.

Suppose there is a linear mapping (transformation) from a mathematical object to another. The derivative of a linear mapping with respect to its argument (i.e. its Jacobian) is constant. Thus, the mapping can be represented and evaluated using the *kernel*, which is the constant Jacobian and, consequently, a tensor. One such example is the transformation between two vector spaces, such as $\boldsymbol{u} = \boldsymbol{A}\boldsymbol{v}$, where $\boldsymbol{u}$ and $\boldsymbol{v}$ are vectors (first-order tensors) and $\boldsymbol{A}$ is a second-order tensor. The perspective of tensors as kernels of linear transformations is widely used to introduce these objects, see for example Holzapfel [1].

The concepts described above seem similar to a linear algebra undergraduate course. Truly, this similarity is the reason behind the wide-spread confusion of tensors with *matrices*. It is fundamental to highlight that matrices are *representational devices*, they are used to tabulate numbers, but there is no constraint to represent a physical quantity. Matrices are useful tools to operate and, above all, for computation, as will be shown in Part II. However, all derivations, from kinematics to differential equations, will be carried out using tensors. Throughout the text, we will refer to matrices with bold letters, such as $\mathbf{u}$ and $\mathbf{A}$, as opposed by bold and italic characters for tensors, $\boldsymbol{u}$ and $\boldsymbol{A}$.

1.2. Matrix representation of tensors

We showed that tensors are arrays of numbers, which preserve physical significance regardless of coordinate system. The components of a tensor can be *tabulated* in matrix format when it is convenient to do so, such as for finite element computations (see Part II). Now, it is convenient to present matrix representation of tensors to completely dissociate one from the other.

To illustrate the representation of tensors in matrix format, we will present examples in three-dimensional space. For spatial coordinates, we will use numbered indices instead of letters; one *possible* correspondence is $x \to 1$, $y \to 2$, and $z \to 3$. Thus, a first-order tensor, such as velocity can be represented by a column matrix as

$$\boldsymbol{v} \xRightarrow[\text{notation}]{\text{matrix}} [v_1 \; v_2 \; v_3]^{\mathrm{T}} \in \mathbb{R}^{3\times 1}; \tag{1}$$

and a second-order tensor as

$$\boldsymbol{\sigma} \xRightarrow[\text{notation}]{\text{matrix}} [\sigma_{11} \;\; \sigma_{12} \;\; \sigma_{13} \;\; \sigma_{21} \;\; \sigma_{22} \;\; \sigma_{23} \;\; \sigma_{31} \;\; \sigma_{32} \;\; \sigma_{33}]^{\mathrm{T}} \in \mathbb{R}^{9\times 1}. \tag{2}$$

Higher-order tensors could have their components tabulated in a column matrix using the same principle—column matrices. Nonetheless, higher-order tensors could also have their components tabulated by two-dimensional matrices, such as $\mathbf{C} \in \mathbb{R}^{9\times 9}$ to represent a fourth-order tensor (with four indices) in three-dimensional space. Henceforth, when a tensor is represented by a matrix, the shape of the latter is immaterial; the only constraint is that eventual algebraic operations between matrix formats are consistent with the original algebraic tensor operations.

1.3. Basis vectors

From this point onward, tensors will materialize as true mathematical objects. It is important to note that all operations and definitions ahead will be carried out in a three-dimensional (3D) space. There is no loss of generality with this assertion, since two-dimensional (2D) space, for instance, is a particular case of the 3D case. As a matter of fact, the overwhelming majority of real-world problems today (*anno Domini* 2026) is tackled in 3D space.

Consider a set of three **linearly independent** and **unitary** vectors (first-order tensors), $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, and $\boldsymbol{e}_3$. Linear independency is extremely important and is defined by the impossibility of representing any element of a linear independent set as a linear combination of the remaining elements [2]. *Unitary*, on the other hand, means those vectors have each a norm (length) of 1. Thus, $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, and $\boldsymbol{e}_3$ form a basis (the elements of the basis being unitary is not a requisite for forming a basis, but it will be quite handy for our purposes).

We can use the previously defined basis to represent any vector using a linear combination of the basis vectors,

$$\boldsymbol{v} = v_1\boldsymbol{e}_1 + v_2\boldsymbol{e}_2 + v_3\boldsymbol{e}_3, \quad v_i \in \mathbb{R}\ \forall i \in \{1, 2, 3\}\,. \tag{3}$$

Conversely, if we define a different basis from vector $\hat{\boldsymbol{e}}_1$, $\hat{\boldsymbol{e}}_2$, and $\hat{\boldsymbol{e}}_3$, vector $\boldsymbol{v}$ can be represented in it too,

$$\boldsymbol{v} = \hat{v}_1\hat{\boldsymbol{e}}_1 + \hat{v}_2\hat{\boldsymbol{e}}_2 + \hat{v}_3\hat{\boldsymbol{e}}_3, \quad \hat{v}_i \in \mathbb{R}\ \forall i \in \{1, 2, 3\}\,. \tag{4}$$

With the important distinction that v_i is probably different than $\hat{v}_i$. This must be a bright moment, for this is a firm grasp onto the very nature of tensors: different basis were defined, different components were computed, and yet $\boldsymbol{v}$ is the same entity in all cases.

The procedure illustrated in this subsection is known as the *basis representation*, and it will be extremely useful ahead, both for derivation and calculation.

1.3.1. Einstein notation

Basis representation is quite cumbersome as written in Eq. (3) and in Eq. (4). Thus, the so-called *Einstein notation* was proposed to compactly represent such set of operation as follows,

$$\boldsymbol{v} = v_1\boldsymbol{e}_1 + v_2\boldsymbol{e}_2 + v_3\boldsymbol{e}_3 = \sum_{i=1}^{3} v_i\boldsymbol{e}_i \xrightarrow[\text{notation}]{\text{Einstein}} v_i\boldsymbol{e}_i. \tag{5}$$

Therefore, Einstein notation is, in plain words, the omission of the sum operator symbol. In some texts, one might observe superscript indices, such as $\boldsymbol{e}^i$, which are usually deployed to indicate *contravariant basis vectors* [1]. However, these are not concerns of this text, and we will stick to *covariant basis vectors*, $\boldsymbol{e}_i$, since we will work on euclidean spaces only.

1.4. Algebraic operations with first-order tensors

There are two basic operations with first-order tensors that are fundamental to constructing vector spaces. Namely,

addition:
$$\boldsymbol{u} + \boldsymbol{v} = u_i\boldsymbol{e}_i + v_i\boldsymbol{e}_i = (u_i + v_i)\,\boldsymbol{e}_i; \tag{6}$$

and

multiplication by scalar: $\lambda \boldsymbol{v} = \lambda \left(v_i \boldsymbol{e}_i \right) = \left(\lambda v_i \right) \boldsymbol{e}_i.$ (7)

It is left to the reader to prove that addition and multiplication by scalar are commutative, i.e. $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$, and associative, $\boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}$ (hint: use Einstein notation and the commutativity and associativity of elements of the field of real numbers, $\mathbb{R}$).

1.4.1. Simple contraction

The simple contraction of two vectors is denoted as

$$\begin{aligned} \boldsymbol{u} \cdot \boldsymbol{v} &= \boldsymbol{v} \cdot \boldsymbol{u} \in \mathbb{R}; \\ \boldsymbol{u} \cdot (\boldsymbol{v} + \boldsymbol{w}) &= \boldsymbol{u} \cdot \boldsymbol{v} + \boldsymbol{u} \cdot \boldsymbol{w} \in \mathbb{R}. \end{aligned} \tag{8}$$

Thus, the simple contraction maps two vectors to a scalar, i.e. $\mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}$. This particular operation is the *projection* of one vector onto the other. In other words, the result, a real number, is the length of the shadow projected onto one vector from the other if the light rays came perpendicular to the vector recipient of the shadow.

In continuum mechanics, we are used to consider problems in an euclidean space, for we neglect space curvature effects such as of general relativity. Thus, it is convenient to choose an orthonormal basis and, consequently, all vectors of the basis are orthogonal to each other. In other words, there is no shadow projected from one vector to another in the elements of the basis. If we plug this property into Eq. (8), we get the widely-known Kronecker's delta δ_{ij},

$$\boldsymbol{e}_i \cdot \boldsymbol{e}_j = \delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

Note that we have not defined any operation law for the simple contraction; we simply equipped it with a geometric sense—projection. However, with Eq. (9) and using Einstein notation as in Eq. (5), we can effectively define a calculation law for Eq. (8),

$$\begin{aligned} \boldsymbol{u} \cdot \boldsymbol{v} &= (u_i \boldsymbol{e}_i) \cdot \left(v_j \boldsymbol{e}_j \right) = (u_1 \boldsymbol{e}_1 + u_2 \boldsymbol{e}_2 + u_3 \boldsymbol{e}_3) \cdot (v_1 \boldsymbol{e}_1 + v_2 \boldsymbol{e}_2 + v_3 \boldsymbol{e}_3) \\ &= u_1 v_1 \boldsymbol{e}_1 \cdot \boldsymbol{e}_1 + u_1 v_2 \boldsymbol{e}_1 \cdot \boldsymbol{e}_2 + u_1 v_3 \boldsymbol{e}_1 \cdot \boldsymbol{e}_3 + u_2 v_1 \boldsymbol{e}_2 \cdot \boldsymbol{e}_1 + u_2 v_2 \boldsymbol{e}_2 \cdot \boldsymbol{e}_2 + u_2 v_3 \boldsymbol{e}_2 \cdot \boldsymbol{e}_3 \\ &\quad + u_3 v_1 \boldsymbol{e}_3 \cdot \boldsymbol{e}_1 + u_3 v_2 \boldsymbol{e}_3 \cdot \boldsymbol{e}_2 + u_3 v_3 \boldsymbol{e}_3 \cdot \boldsymbol{e}_3 = \sum_{i=1}^{3} \sum_{j=1}^{3} u_i v_j \delta_{ij}. \end{aligned} \tag{10}$$

We can use the same principle of omitting the sum operator symbol used to define the

Einstein notation in Eq. (5). Then, Eq. (10) becomes

$$\boldsymbol{u} \cdot \boldsymbol{v} = \sum_{i=1}^{3} \sum_{j=1}^{3} u_i v_j \delta_{ij} = u_i v_j \delta_{ij} \xRightarrow[\text{Eq. (9)}]{\text{From}} \boldsymbol{u} \cdot \boldsymbol{v} = u_i v_i, \tag{11}$$

which is the usual definition of *scalar product* between two vectors.

Contraction naturally appears in the evaluation of physical quantities such as work, power, tractions, etc.

1.4.2. Tensor product

Two vectors (first-order tensors) can be used to create another tensor using the operation known as *tensor product*,

$$\begin{aligned} \boldsymbol{A} &= \boldsymbol{u} \otimes \boldsymbol{v}, \\ \boldsymbol{B} = \boldsymbol{u} \otimes (\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{u} \otimes \boldsymbol{v} + \boldsymbol{u} \otimes \boldsymbol{w} &= \boldsymbol{C} + \boldsymbol{D}. \end{aligned} \tag{12}$$

We can write the vectors involved in the product using Einstein notation, such that

$$\begin{aligned} \boldsymbol{A} = \boldsymbol{u} \otimes \boldsymbol{v} = (u_i \boldsymbol{e}_i) \otimes \left(v_j \boldsymbol{e}_j\right) &= (u_1 \boldsymbol{e}_1 + u_2 \boldsymbol{e}_2 + u_3 \boldsymbol{e}_3) \otimes (v_1 \boldsymbol{e}_1 + v_2 \boldsymbol{e}_2 + v_3 \boldsymbol{e}_3) \\ = u_1 v_1 \boldsymbol{e}_1 \otimes \boldsymbol{e}_1 + u_1 v_2 \boldsymbol{e}_1 \otimes \boldsymbol{e}_2 &+ u_1 v_3 \boldsymbol{e}_1 \otimes \boldsymbol{e}_3 + u_2 v_1 \boldsymbol{e}_2 \otimes \boldsymbol{e}_1 + u_2 v_2 \boldsymbol{e}_2 \otimes \boldsymbol{e}_2 \\ &+ u_2 v_3 \boldsymbol{e}_2 \otimes \boldsymbol{e}_3 + u_3 v_1 \boldsymbol{e}_3 \otimes \boldsymbol{e}_1 + u_3 v_2 \boldsymbol{e}_3 \otimes \boldsymbol{e}_2 + u_3 v_3 \boldsymbol{e}_3 \otimes \boldsymbol{e}_3. \end{aligned} \tag{13}$$

The multiplication pairs between the components of vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ can be rewritten as a generic component $A_{ij} = u_i v_j$,

$$\begin{aligned} \boldsymbol{A} = A_{11} \boldsymbol{e}_1 \otimes \boldsymbol{e}_1 + A_{12} \boldsymbol{e}_1 \otimes \boldsymbol{e}_2 + \cdots &+ A_{32} \boldsymbol{e}_3 \otimes \boldsymbol{e}_2 + A_{33} \boldsymbol{e}_3 \otimes \boldsymbol{e}_3 \\ &= \sum_{i=1}^{3} \sum_{j=1}^{3} A_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j \xRightarrow[\text{notation}]{\text{Einstein}} A_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j. \end{aligned} \tag{14}$$

The components of tensor $\boldsymbol{A}$ in Eq. (14), A_{ij}, have two indices; thus, the tensor product of two first-order tensors generated a second-order tensor. In that sense, the tensor product concatenates two tensors to produce another tensor whose order is equal to the sum of the orders of the previous two tensors.

Exercise 1.1

Prove:

i) $\boldsymbol{u} \otimes \boldsymbol{v} = \boldsymbol{v} \otimes \boldsymbol{u}$ is generally **not true**.

ii) Given $\boldsymbol{u}$, there exist infinitely many $\boldsymbol{v}$ such that $\boldsymbol{u} \otimes \boldsymbol{v} = \boldsymbol{v} \otimes \boldsymbol{u}$.

Remark 1.1

Any tensor order can be constructed using the tensor product and n basis vectors:

$$\text{a generic } n\text{-th order tensor: } \boldsymbol{A} = \underbrace{\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k \otimes \cdots \otimes \boldsymbol{e}_r}_{n} \in \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3. \tag{15}$$

1.5. Second and higher-order tensors

Second-order tensors are extremely useful in continuum mechanics; two examples are: stress and strain in solid mechanics, and heat conductivity in heat transfer. As hinted in Subsubsec. 1.4.2, second-order tensors can be represented by the basis vectors using Einstein notation as

$$\boldsymbol{A} = A_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j, \quad A_{ij} \in \mathbb{R}. \tag{16}$$

One example of particular interest of such tensors is the *second-order identity tensor*, that is defined as

$$\boldsymbol{I} = \delta_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j = \boldsymbol{e}_i \otimes \boldsymbol{e}_i. \tag{17}$$

1.5.1. Transposition of second-order tensors

One of the most important operations with second-order tensors is *transposition*. The transposition of a tensor is denoted as

$$\boldsymbol{A}^{\mathrm{T}} = A_{ij} \boldsymbol{e}_j \otimes \boldsymbol{e}_i = A_{ji} \boldsymbol{e}_i \otimes \boldsymbol{e}_j \implies \left(\boldsymbol{A}^{\mathrm{T}}\right)^{\mathrm{T}} = \boldsymbol{A}, \tag{18}$$

hence, transposing a second-order tensor translates to permuting the basis vectors order or permuting the indices of the components. Moreover, there are two categories of tensors based on how transposition operation behaves with them:

$$1.\ \boldsymbol{A} \text{ is symmetric} \iff \boldsymbol{A} = \boldsymbol{A}^{\mathrm{T}}, \tag{19}$$

$$2.\ \boldsymbol{A} \text{ is skew-symmetric} \iff \boldsymbol{A} = -\boldsymbol{A}^{\mathrm{T}}. \tag{20}$$

As a consequence, it is possible to decompose all second-order tensors into a symmetric parcel, $\mathrm{sym}(\boldsymbol{A})$, and a skew-symmetric parcel, $\mathrm{skew}(\boldsymbol{A})$,

$$\begin{gathered} \boldsymbol{A} = \frac{1}{2}\boldsymbol{A} + \frac{1}{2}\boldsymbol{A} + \frac{1}{2}\boldsymbol{A}^{\mathrm{T}} - \frac{1}{2}\boldsymbol{A}^{\mathrm{T}} = \underbrace{\frac{1}{2}\left(\boldsymbol{A} + \boldsymbol{A}^{\mathrm{T}}\right)}_{\mathrm{sym}(\boldsymbol{A})} + \underbrace{\frac{1}{2}\left(\boldsymbol{A} - \boldsymbol{A}^{\mathrm{T}}\right)}_{\mathrm{skew}(\boldsymbol{A})}; \\ \mathrm{sym}(\boldsymbol{A})^{\mathrm{T}} = \mathrm{sym}(\boldsymbol{A}) \quad \text{and} \quad \mathrm{skew}(\boldsymbol{A})^{\mathrm{T}} = -\,\mathrm{skew}(\boldsymbol{A})\,. \end{gathered} \tag{21}$$

1.5.2. Simple contraction

As discussed in Subsec. 1.1, tensors can be seen as the Jacobian of a mapping between vectors. This mapping is algebraically represented by the contraction of one of the indices of the tensor (usually the second index) with the domain vector. This operation is known as the multiplication of a tensor by a vector or the contraction of a tensor with a vector. Such contraction can be represented in Einstein notation as

$$\boldsymbol{v} = v_i \boldsymbol{e}_i = \boldsymbol{A}\boldsymbol{u} = \boldsymbol{A} \cdot \boldsymbol{u} = \left(A_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j\right) \cdot \left(u_k \boldsymbol{e}_k\right) = A_{ij} u_k \left(\boldsymbol{e}_i \otimes \boldsymbol{e}_j \cdot \boldsymbol{e}_k\right). \tag{22}$$

Even though a contraction is carried out (between the second index of the second-order tensor with the single index of the vector), the dot $\cdot$ is omitted to simplify notation. This omission is conventionally used for tensor-vector multiplication and tensor-tensor multiplication only. The component u_k was *carried to the front* due to the distributivity of the contraction operation, see Eq. (8), it is a good exercise for the reader to demonstrate the consistency of such *trickery* by writing all components explicitly.

Using the orthogonality of the conventionalized basis, the contraction in Eq. (22) can be simplified to the Kronecker's delta,

$$\boldsymbol{A}\boldsymbol{u} = A_{ij} u_k \delta_{jk} \boldsymbol{e}_i = A_{ij} u_j \boldsymbol{e}_i = v_i \boldsymbol{e}_i \implies v_i = A_{ij} u_j = A_{i1} u_1 + A_{i2} u_2 + A_{i3} u_3. \tag{23}$$

Note that the index k of the component u_k was changed to j, i.e. $u_k \to u_j$. This is justified by the fact that the Kronecker's delta δ_{jk} is equal to zero for all $j \neq k$, thus, we simply chose j in spite of k. Observe as well that the index j becomes isolated in the expression $v_i = A_{ij} u_j$, in the sense that there is no correspondence in the right-hand side. In fact, this index is there just to indicate a sum, as clarified in the final step of Eq. (23). Therefore, j, in this scenario, is the example of a *dummy index*.

Exercise 1.2

Prove that $\boldsymbol{v} \cdot (\boldsymbol{A}\boldsymbol{u}) = \left(\boldsymbol{A}^{\mathrm{T}} \boldsymbol{v}\right) \cdot \boldsymbol{u} \in \mathbb{R}$.

The multiplication between two tensors is given by the contraction of the second index of the first tensor with the first index of the second tensor,

$$\begin{aligned} \boldsymbol{A}\boldsymbol{B} = \left(A_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j\right) \cdot \left(B_{kl} \boldsymbol{e}_k \otimes \boldsymbol{e}_l\right) = A_{ij} B_{kl} \left(\boldsymbol{e}_i \otimes \boldsymbol{e}_j \cdot \boldsymbol{e}_k \otimes \boldsymbol{e}_l\right) = A_{ij} B_{kl} \delta_{jk} \boldsymbol{e}_i \otimes \boldsymbol{e}_l \\ = A_{ij} B_{jl} \boldsymbol{e}_i \otimes \boldsymbol{e}_l = \left(A_{i1} B_{1l} + A_{i2} B_{2l} + A_{i3} B_{3l}\right) \boldsymbol{e}_i \otimes \boldsymbol{e}_l. \end{aligned} \tag{24}$$

In Eq. (23), two indices were contracted (i.e. one made a dummy index and the other eliminated), the second index of the tensor $\boldsymbol{A}$ and the only index of the vector $\boldsymbol{u}$. Thus, the first index of $\boldsymbol{A}$ is the only non dummy index left. In Eq. (24), the same pattern appeared—the second index of the tensor $\boldsymbol{A}$ contracted with the first index of $\boldsymbol{B}$; however, two indices remained: the first index of $\boldsymbol{A}$ and the second index of $\boldsymbol{B}$. In summary, it becomes clear that a simple contraction, denoted by the single dot, $\cdot$, or the omission of it, takes away two *free indices* away from the original set of indices.

Exercise 1.3

i) Prove that $\boldsymbol{u} = \boldsymbol{I}\boldsymbol{u}$, where $\boldsymbol{I}$ is the second-order identity tensor.

ii) If $\boldsymbol{AB} = \boldsymbol{BA}$, it is said that $\boldsymbol{A}$ and $\boldsymbol{B}$ *commute*. Prove that, in general, two generic tensors $\boldsymbol{A}$ and $\boldsymbol{B}$ do **not** commute; but any tensor $\boldsymbol{A}$ commutes with the second-order identity tensor $\boldsymbol{I}$.

iii) Prove that $(\boldsymbol{AB})^{\mathrm{T}} = \boldsymbol{B}^{\mathrm{T}}\boldsymbol{A}^{\mathrm{T}}$.

1.5.3. Inverse of second-order tensors

In Eq. (22), second-order tensors were shown as the Jacobian of linear mapping between vector spaces. If a mapping $f : \boldsymbol{u} \mapsto \boldsymbol{v}$ is a *bijection*, i.e. for each vector $\boldsymbol{u}$ in the domain there is a single corresponding vector $\boldsymbol{v} = f(\boldsymbol{u})$, f is *invertible*, whose inverse is denoted as f^{-1}, such that $f^{-1}\left(f(\boldsymbol{u})\right) = \boldsymbol{u}$. In particular, for linear mappings,

$$\boldsymbol{v} = f(\boldsymbol{u}) = \boldsymbol{A}\boldsymbol{u} \text{ is a bijection } \implies \boldsymbol{u} = f^{-1}\left(f(\boldsymbol{u})\right) = f^{-1}(\boldsymbol{A}\boldsymbol{u}) = \boldsymbol{A}^{-1}\boldsymbol{A}\boldsymbol{u}. \tag{25}$$

We might rearrange Eq. (25) to

$$\boldsymbol{u} - \boldsymbol{A}^{-1}\boldsymbol{A}\boldsymbol{u} = \left[\boldsymbol{I} - \boldsymbol{A}^{-1}\boldsymbol{A}\right]\boldsymbol{u} = \boldsymbol{0} \;\; \forall \boldsymbol{u} \in \mathbb{R}^3 \implies \boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}. \tag{26}$$

Therefore, if $\boldsymbol{A}$ is invertible, the *inverse of* $\boldsymbol{A}$, $\boldsymbol{A}^{-1}$, is given by,

$$\text{if } \boldsymbol{A} \text{ is invertible, } \boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I} \implies A^{-1}_{ik}A_{kj} = A_{ik}A^{-1}_{kj} = \delta_{ij}. \tag{27}$$

Exercise 1.4

i) Prove that $\boldsymbol{A}$ and $\boldsymbol{A}^{-1}$ commute. Hint: use the inverse mapping f^{-1} definition to get $\boldsymbol{v}$ instead of $\boldsymbol{u}$.

ii) Prove that $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.

1.5.4. Double contraction

Double contraction is a common operation between a second-order tensor and: (i) another second-order tensor, to evaluate the energy or power; (ii) a third-order tensor, to evaluate rotation axle vectors; (iii) a fourth-order tensor, to relate stress and strain tensors. We will first define the double contraction using a n-th order tensor $\boldsymbol{A}$ (see Eq. 15) and a

m-th order tensor $\boldsymbol{B}$,

$$\boldsymbol{A} : \boldsymbol{B} = \left(A_{ij\ldots trs}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_t \otimes \boldsymbol{e}_r \otimes \boldsymbol{e}_s\right) : \left(B_{\alpha\beta\gamma\ldots\xi}\boldsymbol{e}_\alpha \otimes \boldsymbol{e}_\beta \otimes \boldsymbol{e}_\gamma \otimes \cdots \otimes \boldsymbol{e}_\xi\right)$$

$$= A_{ij\ldots trs}B_{\alpha\beta\gamma\ldots\xi}\delta_{r\alpha}\delta_{s\beta}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_t \otimes \boldsymbol{e}_\gamma \otimes \cdots \otimes \boldsymbol{e}_\xi$$

$$= \underbrace{A_{ij\ldots trs}B_{rs\gamma\ldots\xi}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_t \otimes \boldsymbol{e}_\gamma \otimes \cdots \otimes \boldsymbol{e}_\xi}_{(n+m-4)\text{-th order tensor}}. \tag{28}$$

This operation can be defined differently depending on the text and physical needs. However, for this text, the double contraction is defined in words as: *the contraction of the two last indices of the first tensor with the first two indices of the second tensor.* Another interesting fact of Eq. (28) is that, in combination with Eq. (24), each dot in a contraction operation takes away two indices of the original set of free indices. Thus, in Eq. (28), $n+m$ free indices were originally presented, but the final result is a tensor of $(n+m-4)$-th order.

Three particularly useful examples of double contractions are: (i) the contraction of two second-order tensors to yield a scalar,

$$\boldsymbol{P} : \dot{\boldsymbol{F}} = P_{ij}\dot{F}_{ij} \in \mathbb{R} \;\rightarrow \text{power volumetric density [J/sm}^3\text{]}, \tag{29}$$

where J is the energy unit joule, and s is time unit second; (ii) the contraction of a fourth-order tensor with a second-order one,

$$\mathbb{C} : \boldsymbol{\varepsilon} = \mathbb{C}_{ijkl}\varepsilon_{kl}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \;\rightarrow \text{stress [Pa]}; \tag{30}$$

finally, (iii) the contraction of a second-order tensor with the result of the contraction a fourth and another second-order tensors,

$$\boldsymbol{F} : \mathbb{C} : \boldsymbol{F} = F_{ij}\mathbb{C}_{ijkl}F_{kl} \in \mathbb{R} \;\rightarrow \text{energy volumetric density [J/m}^3\text{]}. \tag{31}$$

Exercise 1.5

i) Demonstrate that $\boldsymbol{S} : \boldsymbol{W} = 0$, where $\boldsymbol{S}$ is a symmetric second-order tensor and $\boldsymbol{W}$ is a skew-symmetric second-order tensor.

ii) Given two second-order tensors, $\boldsymbol{A}$ and $\boldsymbol{B}$, prove that $\boldsymbol{A} : \boldsymbol{B}$ can be evaluated using the symmetric and skew-symmetric parcels of each one of the two tensors.

iii) Prove that $\boldsymbol{A} : \boldsymbol{B} = \boldsymbol{B} : \boldsymbol{A} = \boldsymbol{A}^{\mathrm{T}} : \boldsymbol{B}^{\mathrm{T}} = \boldsymbol{B}^{\mathrm{T}} : \boldsymbol{A}^{\mathrm{T}}$.

iv) Prove that $\boldsymbol{AB} : \boldsymbol{C} = \boldsymbol{ABC}^{\mathrm{T}} : \boldsymbol{I}$.

Exercise 1.6

Show that

$$\boldsymbol{A} : \boldsymbol{B}\boldsymbol{C} = \boldsymbol{A}\boldsymbol{C}^{\mathrm{T}} : \boldsymbol{B}. \tag{32}$$

1.5.5. An important third-order tensor—the permutation tensor

The permutation tensor is a third-order tensor defined as

$$\boldsymbol{\varepsilon} = \varepsilon_{ijk}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \boldsymbol{e}_k; \quad \begin{cases} \varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312} = 1\text{: } \textbf{even} \text{ permutations of } (i,j,k), \\ \varepsilon_{132} = \varepsilon_{213} = \varepsilon_{321} = -1\text{: } \textbf{odd} \text{ permutations of } (i,j,k), \\ 0 \text{ otherwise.} \end{cases} \tag{33}$$

Even and odd permutations refers to the number of times the tuple of indices (i, j, k) must be permuted to get a particular combination. For instance, $(123) \to (312)$ corresponds to at least 2 permutations, thus an even number of permutations; see: $(123) \to (132) \to (312)$. The components of the permutation, or Levi-Civita, tensor present the following properties [1],

$$\begin{aligned} \textbf{even} \text{ permutations: } & \varepsilon_{ijk} = \varepsilon_{jki} = \varepsilon_{kij}; \\ \textbf{odd} \text{ permutations: } & \varepsilon_{ijk} = -\varepsilon_{jik} = -\varepsilon_{ikj}. \end{aligned} \tag{34}$$

The permutation tensor is useful to evaluate the vector $\boldsymbol{w}$ that is simultaneously orthogonal to vectors $\boldsymbol{u}$ and $\boldsymbol{v}$, i.e. the vector product between $\boldsymbol{u}$ and $\boldsymbol{v}$,

$$\boldsymbol{w} = \boldsymbol{u} \times \boldsymbol{v} = \varepsilon_{ijk}u_i v_j \boldsymbol{e}_k = (\boldsymbol{u} \otimes \boldsymbol{v}) : \boldsymbol{\varepsilon} = \varepsilon_{ijk}u_j v_k \boldsymbol{e}_i = \boldsymbol{\varepsilon} : (\boldsymbol{u} \otimes \boldsymbol{v})\,. \tag{35}$$

Exercise 1.7

i) Prove that $\boldsymbol{w} = \boldsymbol{u} \times \boldsymbol{v} = \boldsymbol{\varepsilon} : (\boldsymbol{u} \otimes \boldsymbol{v})$.

ii) Given the vector $\boldsymbol{\varphi}$, prove that $\boldsymbol{W} = \boldsymbol{\varepsilon} \cdot \boldsymbol{\varphi}$ is skew-symmetric.

iii) Given $\boldsymbol{W} = \boldsymbol{\varepsilon} \cdot \boldsymbol{\varphi}$, prove that $\boldsymbol{W}\boldsymbol{v} = \boldsymbol{\varphi} \times \boldsymbol{v}$; and, consequently that $(\boldsymbol{\varepsilon} \cdot \boldsymbol{\varphi}) \cdot \boldsymbol{\varphi} = \boldsymbol{0}$.

iv) Prove that $\boldsymbol{\varepsilon} : \boldsymbol{\varepsilon} \cdot \boldsymbol{\varphi} = \boldsymbol{\varphi}$.

Remark 1.2

As a skew-symmetric tensor $\boldsymbol{W}$ has only 3 free components, it can be represented by a vector, which is known as the *axial vector*,

$$\boldsymbol{\varphi} = \operatorname{axl}(\boldsymbol{W}) = \boldsymbol{\varepsilon} : \boldsymbol{W} \xRightarrow[\text{the canonical basis}]{\text{Matrix notation in}} [\boldsymbol{W}] = \begin{bmatrix} 0 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 0 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 0 \end{bmatrix}. \tag{36}$$

1.5.6. Trace and determinant of second-order tensors

The trace of a second-order tensor is a mapping $\operatorname{tr}() : \mathbb{R}^{3\times3} \mapsto \mathbb{R}$ defined as

$$\operatorname{tr}(\boldsymbol{A}) = \boldsymbol{A}_{ii}, \tag{37}$$

in other words, the sum of all components that have the first index equal to the second. The trace will play a major role in computing and manipulating stress tensors.

Exercise 1.8

i) Prove that $\operatorname{tr}(\boldsymbol{A}) = \boldsymbol{I} : \boldsymbol{A}$.

ii) Prove that $\operatorname{tr}\left(\boldsymbol{A}^{\mathrm{T}}\right) = \operatorname{tr}(\boldsymbol{A})$.

iii) Prove that the trace is a *linear operator*, i.e. $\operatorname{tr}(\boldsymbol{A} + \lambda\boldsymbol{B}) = \operatorname{tr}(\boldsymbol{A}) + \lambda\operatorname{tr}(\boldsymbol{B}) \quad \forall\lambda \in \mathbb{R}$.

iv) Prove that $\boldsymbol{A} : \boldsymbol{B} = \operatorname{tr}\left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{B}\right)$.

The determinant of a second-order is a mapping $\det() : \mathbb{R}^{3\times3} \mapsto \mathbb{R}$ defined such that

$$\begin{aligned} \det(\boldsymbol{AB}) &= \det(\boldsymbol{A})\det(\boldsymbol{B}), \\ \det\left(\boldsymbol{A}^{\mathrm{T}}\right) &= \det(\boldsymbol{A}). \end{aligned} \tag{38}$$

In particular, for our needs, we can formally define the evaluation of the determinant of a second-order tensor using the permutation tensor components,

$$\det(\boldsymbol{A}) = \varepsilon_{ijk} A_{1i} A_{2j} A_{3k}. \tag{39}$$

Historically, the determinant, although seemingly a mysterious operation, is intrinsically linked to areas and volumes. Determinants are operations formally defined to calculate the *measure* of a space through differential forms. This topic is left to another opportunity where differential geometry and measure theory might be subjects. However, as a hint to the reader, the determinant is used to compute differential forms in curvilinear coordinate systems, such as area and volume differentials in cylindrical or spherical coordinate systems.

Exercise 1.9

i) Prove that $\det(\lambda \boldsymbol{A}) = \lambda^3 \det(\boldsymbol{A}), \quad \forall \lambda \in \mathbb{R}$.

ii) Prove that $\det\left(\boldsymbol{A}^{-1}\right) = \det(\boldsymbol{A})^{-1}$ if and only if $\boldsymbol{A}$ is invertible.

1.5.7. Orthogonal second-order tensors

An orthogonal second-order tensor is defined as

$$\boldsymbol{A} \in \text{Orth if } \boldsymbol{A}^{-1} = \boldsymbol{A}^{\mathrm{T}}. \tag{40}$$

In particular, if the determinant of an orthogonal tensor is equal to 1, it is said to be a *proper orthogonal tensor*, Orth^+, or a member of the group of rotations in 3D space, SO^3. As hinted before, the geometric interpretation of such tensors is a rotation about an axle. If the determinant is equal to -1, the tensor is said be an *improper orthogonal tensor*, Orth^-, whose geometric interpretation is a reflection on a plane.

Given that there are two types of orthogonal tensors, we will give explicit formulas for them. One must note that these special tensors can be fully defined by vectors—a rotation about an axle (represented by a vector) and a reflection on a plane (given by the normal vector of the plane).

Any 3D rotation tensor can be given by the Euler-Rodrigues formula [3, 4],

$$\begin{aligned} \boldsymbol{R}(\boldsymbol{\varphi}) = \cos\varphi \boldsymbol{I} + \frac{\sin\varphi}{\varphi}\boldsymbol{W} + \frac{1-\cos\varphi}{\varphi^2}\boldsymbol{\varphi}\otimes\boldsymbol{\varphi}; \\ \boldsymbol{W} = \boldsymbol{\varepsilon}\cdot\boldsymbol{\varphi},\ \varphi = \|\boldsymbol{\varphi}\| = \sqrt{\boldsymbol{\varphi}\cdot\boldsymbol{\varphi}}; \end{aligned} \tag{41}$$

where $\boldsymbol{\varphi}$ is a vector colinear to the rotation axle and whose euclidean norm, $\varphi = \|\boldsymbol{\varphi}\|$, equals the rotation angle in radians. Rotation, then, becomes a linear transformation,

$$\boldsymbol{v} = \boldsymbol{R}(\boldsymbol{\varphi})\,\boldsymbol{u}; \tag{42}$$

whose explanation in words is: *$\boldsymbol{v}$ is the vector $\boldsymbol{u}$ rotated about the axle $\boldsymbol{\varphi}$ by $\|\boldsymbol{\varphi}\|$ radians.*

Conversely, any reflection tensor is given by

$$\boldsymbol{M}(\boldsymbol{n}) = \boldsymbol{I} - 2\boldsymbol{n}\otimes\boldsymbol{n}; \quad \boldsymbol{n}\cdot\boldsymbol{n} = 1, \tag{43}$$

where $\boldsymbol{n}$ is a unitary vector normal to the reflection plane. As is the case of rotations, reflection is also a linear transformation,

$$\boldsymbol{v} = \boldsymbol{M}(\boldsymbol{n})\,\boldsymbol{u}; \tag{44}$$

whose explanation in words is: *$\boldsymbol{v}$ is the vector $\boldsymbol{u}$ reflected through the plane whose normal vector is $\boldsymbol{n}$.*

1.5.8. Rotation of tensors

As discussed before, to rotate a vector about an axle $\boldsymbol{\varphi}$, one should simply do $\boldsymbol{v} = \boldsymbol{R}\boldsymbol{u}$. If a tensor is to be rotated, the vectors of the tensor products must be individually rotated,

$$\boldsymbol{A} = \boldsymbol{u}_i \otimes \boldsymbol{u}_j \otimes \cdots \otimes \boldsymbol{u}_s \xRightarrow[\text{by } \boldsymbol{R}]{\text{Rotation}} \bar{\boldsymbol{A}} = \boldsymbol{v}_i \otimes \boldsymbol{v}_j \otimes \dots \boldsymbol{v}_s = (\boldsymbol{R}\boldsymbol{u}_i) \otimes \left(\boldsymbol{R}\boldsymbol{u}_j\right) \otimes \cdots \otimes (\boldsymbol{R}\boldsymbol{u}_s)\,; \tag{45}$$

which can be generically extended to Einstein notation in the canonical basis,

$$\begin{aligned} \boldsymbol{A} = A_{ij\dots s}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_s \xRightarrow[\text{by } \boldsymbol{R}]{\text{Rotation}} \bar{\boldsymbol{A}} &= \bar{A}_{ij\dots s}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_s \\ \bar{A}_{ij\dots s} &= \boldsymbol{R}_{i\alpha}\boldsymbol{R}_{j\beta}\dots\boldsymbol{R}_{s\xi}A_{\alpha\beta\dots\xi}. \end{aligned} \tag{46}$$

1.5.9. Coordinates transformation

It is not uncommon for people to mistake rotation for coordinate transformation and vice-versa. The rotation operation results in the components of the rotated tensor in the same system of coordinates; coordinate transformation is an operation to get the components of a tensor in a different system of coordinates.

Suppose there is a system of coordinates, whose basis vectors are $\left\{\hat{\boldsymbol{e}}_i,\ \ i \in \{1,2,3\}\right\}$, that is rotated with reference to another system of coordinates whose set of basis vectors is $\left\{\boldsymbol{e}_i,\ \ i \in \{1,2,3\}\right\}$. This rotation is expressed by

$$\hat{\boldsymbol{e}}_i = \boldsymbol{Q}\boldsymbol{e}_i,\ \ \forall i \in \{1,2,3\}\,;\ \boldsymbol{Q} \in \mathrm{SO}^3. \tag{47}$$

Suppose, now, that we have a vector $\boldsymbol{v}$. To evaluate its components in the new (rotated) system of coordinates, we need to project the vector onto the basis of the said system,

$$\boldsymbol{v} = v_i\boldsymbol{e}_i \xRightarrow[\text{rotated coordinates}]{\text{Components in}} \boldsymbol{v} = \hat{v}_i\hat{\boldsymbol{e}}_i,\ \ \hat{v}_i = \boldsymbol{v}\cdot\hat{\boldsymbol{e}}_i = \left(v_j\boldsymbol{e}_j\right)\cdot\hat{\boldsymbol{e}}_i = v_j\boldsymbol{e}_j\cdot\boldsymbol{Q}\boldsymbol{e}_i. \tag{48}$$

We might write the transformation tensor $\boldsymbol{Q}$ using Einstein notation in the basis of the original system, then,

$$\hat{v}_i = v_j\boldsymbol{e}_j\cdot\left(Q_{kl}\boldsymbol{e}_k\otimes\boldsymbol{e}_l\right)\boldsymbol{e}_i = v_jQ_{kl}\delta_{jk}\delta_{li} = v_jQ_{ji}. \tag{49}$$

The same procedure can be carried out to a second-order tensor,

$$\hat{A}_{ij} = \left(\left(\boldsymbol{A}_{kl}\boldsymbol{e}_k\otimes\boldsymbol{e}_l\right)\cdot\hat{\boldsymbol{e}}_j\right)\cdot\hat{\boldsymbol{e}}_i = \left(A_{kl}Q_{lj}\boldsymbol{e}_k\right)\cdot\boldsymbol{Q}\boldsymbol{e}_i = A_{kl}Q_{lj}Q_{ki}. \tag{50}$$

A particular useful category of second-order tensor is an *isotropic tensor*, which must follow the rule

$$A_{ij} = A_{kl} Q_{lj} Q_{ki}. \tag{51}$$

Exercise 1.10

i) If the squared euclidean norm of the vector $\boldsymbol{u}$ is $\|\boldsymbol{u}\|^2 = \boldsymbol{u} \cdot \boldsymbol{u}$, prove that given a rotation tensor $\boldsymbol{R}$, the rotated vector $\boldsymbol{v} = \boldsymbol{R}\boldsymbol{u}$ has the same euclidean norm as $\boldsymbol{u}$.

ii) Prove that a reflected vector $\boldsymbol{v} = \boldsymbol{M}\boldsymbol{u}$ by a reflection tensor $\boldsymbol{M}$ has the same euclidean norm as $\boldsymbol{u}$.

iii) Show that $\boldsymbol{R}(\boldsymbol{\varphi})\,\boldsymbol{\varphi} = \boldsymbol{\varphi}$.

Exercise 1.11

i) Show that any vector $\boldsymbol{v}$ perpendicular to $\boldsymbol{n}$, $\boldsymbol{v} \cdot \boldsymbol{n} = 0$, is not affected by $\boldsymbol{M}(\boldsymbol{n})$, i.e. $\boldsymbol{v} = \boldsymbol{M}(\boldsymbol{n})\,\boldsymbol{v}$.

ii) Show that $\det\left(\boldsymbol{R}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{R}\right) = \det(\boldsymbol{A})$.

iii) Show that $\operatorname{tr}\left(\boldsymbol{R}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{R}\right) = \operatorname{tr}(\boldsymbol{A})$. Hint: use the identity $\operatorname{tr}\left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{B}\right) = \boldsymbol{A} : \boldsymbol{B} = \boldsymbol{B} : \boldsymbol{A}$.

1.5.10. Eigenvectors and eigenvalues

A second-order tensor $\boldsymbol{A}$ has three different eigenvectors, denoted by $\boldsymbol{n}$, and three (not necessarily different) eigenvalues, denoted by λ, such that

$$\boldsymbol{A}\boldsymbol{n} = \lambda\boldsymbol{n}, \;\; \lambda \in \mathbb{R} \implies [\boldsymbol{A} - \lambda\boldsymbol{I}]\,\boldsymbol{n} = \boldsymbol{0} \implies \det(\boldsymbol{A} - \lambda\boldsymbol{I}) = 0. \tag{52}$$

For second-order tensors in 3D space, the null-determinant condition particularizes to a cubic polynomial equation,

$$\begin{aligned}
(A_{11} - \lambda)(A_{22} - \lambda)(A_{33} - \lambda) + A_{12}A_{23}A_{31} + A_{13}A_{21}A_{32} - (A_{11} - \lambda)A_{23}A_{32} \\
-A_{12}A_{21}(A_{33} - \lambda) - A_{13}(A_{22} - \lambda)A_{31} = \\
-\lambda^3 + (A_{11} + A_{22} + A_{33})\lambda^2 - (A_{11}A_{22} + A_{11}A_{33} + A_{22}A_{33} - A_{23}A_{32} - A_{12}A_{21} - A_{13}A_{31})\lambda \\
+A_{11}A_{22}A_{33} + A_{12}A_{23}A_{31} + A_{13}A_{21}A_{32} - A_{11}A_{23}A_{32} - A_{12}A_{21}A_{33} - A_{13}A_{22}A_{31} \\
-\lambda^3 + I_1\lambda^2 - I_2\lambda + I_3 = 0,
\end{aligned} \tag{53}$$

which is known as the *characteristic polynomial* of $\boldsymbol{A}$. By inspection of Eq. (53), it is straightforward to show that

$$
\begin{aligned}
I_1(\boldsymbol{A}) &= \operatorname{tr}(\boldsymbol{A}), \\
I_2(\boldsymbol{A}) &= \frac{1}{2}\left[\operatorname{tr}(\boldsymbol{A})^2 - \operatorname{tr}\left(\boldsymbol{A}^2\right)\right], \\
I_3(\boldsymbol{A}) &= \det(\boldsymbol{A}).
\end{aligned} \tag{54}
$$

Remark 1.3

The Cayley-Hamilton theorem states that a tensor $\boldsymbol{A}$ satisfies its own characteristic equation (polynomial), such that

$$
-\boldsymbol{A}^3 + I_1\boldsymbol{A}^2 - I_2\boldsymbol{A} + I_3\boldsymbol{I} = \boldsymbol{0}, \tag{55}
$$

where $\boldsymbol{0}$ is the second-order null tensor. This theorem is quite useful to evaluating tensor functions and tensor derivatives.

Exercise 1.12

Given $\boldsymbol{R} \in \mathrm{SO}^3$, prove that $I_1\left(\boldsymbol{R}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{R}\right) = I_1(\boldsymbol{A})$, $I_2\left(\boldsymbol{R}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{R}\right) = I_2(\boldsymbol{A})$, and $I_3\left(\boldsymbol{R}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{R}\right) = I_3(\boldsymbol{A})$.

If $I_i\left(\boldsymbol{R}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{A}\right) = I_i(\boldsymbol{A}) \quad \forall i \in \{1,2,3\}$ and $\forall \boldsymbol{R} \in \mathrm{SO}^3$, I_i is said to be an *invariant*, since its value is invariant to a rotation of the coordinate system. Consequently, the eigenvalues of a tensor $\boldsymbol{R}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{R}$ are equal to the eigenvalues of $\boldsymbol{A}$ for any $\boldsymbol{R} \in \mathrm{SO}^3$; this transformation by $\boldsymbol{R}$ is also known as a *similarity transformation*, since the eigenvalues are preserved. Similarity transformations will play a major role in the principal direction of stretch in the next section of kinematics.

Remark 1.4

Any symmetric tensor $\boldsymbol{A}$, $\boldsymbol{A} = \boldsymbol{A}^{\mathrm{T}}$, can be represented by its *spectral decomposition*,

$$
\boldsymbol{A} = \lambda_i \boldsymbol{n}_i \otimes \boldsymbol{n}_i; \quad \boldsymbol{A}\boldsymbol{n}_i = \lambda_i \boldsymbol{n}_i, \quad \|\boldsymbol{n}_i\|_2 = 1. \tag{56}
$$

Proof:

$$
\boldsymbol{A} = \boldsymbol{A}\boldsymbol{I} = \boldsymbol{A}(\boldsymbol{n}_i \otimes \boldsymbol{n}_i) = \lambda_i \boldsymbol{n}_i \otimes \boldsymbol{n}_i. \tag{57}
$$

The eigenvectors of a tensor can be used to form a basis for a vector space $\mathcal{V}$. The vector space generated by the span of the basis of a second-order tensor $\boldsymbol{A}$ is known as the *eigenspace of* $\boldsymbol{A}$, such that $\mathcal{V} = \operatorname{eigspace}(\boldsymbol{A})$. Eigenspaces are important to show commutativity, when two tensors commute through the simple contraction, and, therefore, to show that the stress response of a material is isotropic.

Exercise 1.13

1. Use identity $\boldsymbol{n}_i \cdot \boldsymbol{A}\boldsymbol{n}_j = \left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{n}_i\right) \cdot \boldsymbol{n}_j$ to prove that the eigenvectors of a symmetric tensor $\boldsymbol{A}$, such that $\boldsymbol{A} = \boldsymbol{A}^{\mathrm{T}}$, are orthogonal one to another.
2. Use the spectral decomposition to show that two symmetric tensors commute if, and only if, they share the same eigenspace.

Challenge 1.1

Prove that, if $\boldsymbol{A} = \boldsymbol{A}^{\mathrm{T}}$, $\boldsymbol{u} \cdot (\boldsymbol{A}\boldsymbol{u}) > 0$ if and only if the eigenvalues of $\boldsymbol{A}$ are strictly positive. Then $\boldsymbol{A}$ is said to be positive-definite, $\boldsymbol{A}$ pos. def., i.e.

$$\boldsymbol{A} \text{ pos. def.} \iff \boldsymbol{u} \cdot \boldsymbol{A}\boldsymbol{u} > 0 \;\; \forall \boldsymbol{u} \in \mathbb{R}^3. \tag{58}$$

1.6. Inner product

Inner product is a map from a vector (or function) space V to an element of the field (a real scalar in our case),

$$\langle \cdot, \cdot \rangle : V \times V \mapsto \mathbb{R}, \tag{59}$$

that must satisfy the following properties,

$$\begin{aligned} \text{commutativity:} \;\; & \langle \boldsymbol{u}, \boldsymbol{v} \rangle = \langle \boldsymbol{v}, \boldsymbol{u} \rangle \, ; \\ \text{linearity:} \;\; & \langle \boldsymbol{u}, \alpha\boldsymbol{v} + \beta\boldsymbol{w} \rangle = \alpha \langle \boldsymbol{u}, \boldsymbol{v} \rangle + \beta \langle \boldsymbol{u}, \boldsymbol{w} \rangle \, , \;\; \forall \alpha, \beta \in \mathbb{R}; \\ \text{positive-definiteness:} \;\; & \langle \boldsymbol{u}, \boldsymbol{u} \rangle = 0 \iff \boldsymbol{u} = \boldsymbol{0} \text{ and } \langle \boldsymbol{u}, \boldsymbol{u} \rangle > 0 \;\; \forall \boldsymbol{u} \neq \boldsymbol{0}. \end{aligned} \tag{60}$$

The space endowed with an inner product, such as V, is said to be an *inner product space.*

Inner products are extensively used in continuum mechanics to evaluate norm of tensor entities, to evaluate orthogonality between tensors, and to evaluate energy-related metrics. Inner products between function spaces will be absolutely necessary for the finite element method, in Part II.

Exercise 1.14

i) Prove that the simple contraction between vectors, $\boldsymbol{u} \cdot \boldsymbol{v}$, is an inner product.

ii) Prove that the double contraction between second-order tensors, $\boldsymbol{A} : \boldsymbol{B}$, is an inner product.

Challenge 1.2

Given a mapping defined as $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \boldsymbol{u} \cdot (\boldsymbol{A}\boldsymbol{v})$, where $\cdot$ is the simple contraction and $\boldsymbol{A}$ is a second-order tensor, state the properties required of $\boldsymbol{A}$ for that mapping to be an inner product.

Remark 1.5

A *quadratic form* of a vector $\boldsymbol{u}$ is defined as $\boldsymbol{u} \cdot (\boldsymbol{A}\boldsymbol{u})$ ($\boldsymbol{A}$ is a second-order tensor), and a quadratic form of a second-order tensor $\boldsymbol{B}$ is defined as $\boldsymbol{B} : \mathbb{C} : \boldsymbol{B}$ ($\mathbb{C}$ is a fourth-order tensor). Both are extremely important to asserting if a point is a minimum, a maximum, or a saddle-point in an optimization problem. Thus, when accounting for energy or power, they are necessary to prove that a BVP has a unique solution.

2. Calculus with tensors

2.1. Tensor-valued functions of tensor arguments

Sec. 1 was dedicated to defining what a tensor is and how to algebrically operate with these mathematical objects. We briefly showed some mappings before, such as rotations, projections, and inner products. Now, it is appropriate to show proper *functions* of tensors. Hence, we will commence by listing some useful function categories, defined by the argument (the object in the domain, i.e. the input) and by the image (the object in the counter-domain, i.e. the output). We will denote a generic function in this subsection by the symbol f.

2.1.1. Functions of scalar argument

This kind of functions has $\mathbb{R}$ as domain, i.e. they receive a scalar as argument. They can be further subdivided according to their output,

1. Scalar-valued: $f : \mathbb{R} \mapsto \mathbb{R}$. Example: energy density in time, $\psi = \psi(t) \in \mathbb{R}$.

2. Vector-valued: $f : \mathbb{R} \mapsto \mathbb{R}^3$. Example: velocity in time, $\boldsymbol{v} = \boldsymbol{v}(t) \in \mathbb{R}^3$.

3. Tensor-valued: $f : \mathbb{R} \mapsto \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3$. Example: strain in time, $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}(t) \in \mathbb{R}^3 \otimes \mathbb{R}^3$.

2.1.2. Functions of vector argument

This kind of functions has $\mathbb{R}^3$ as domain, i.e. they receive a vector as argument. They can be further subdivided according to their output,

1. Scalar-valued: $f : \mathbb{R}^3 \mapsto \mathbb{R}$. Example: the norm of a vector, $\|\boldsymbol{v}\|_2 = \sqrt{\boldsymbol{v} \cdot \boldsymbol{v}} \in \mathbb{R}$.

2. Vector-valued: $f : \mathbb{R}^3 \mapsto \mathbb{R}^3$. Example: linear transformation of a vector, such as rotation, $\boldsymbol{v}(\boldsymbol{u}) = \boldsymbol{R}\boldsymbol{u} \in \mathbb{R}^3$.

3. Tensor-valued: $f : \mathbb{R}^3 \mapsto \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3$. Example: the skew-symmetric tensor given an axial vector, $\boldsymbol{W}(\boldsymbol{u}) = \boldsymbol{\varepsilon} \cdot \boldsymbol{u} \in \mathbb{R}^3 \otimes \mathbb{R}^3$, where $\boldsymbol{\varepsilon}$ is the third-order permutation tensor.

2.1.3. *Functions of tensor argument*

This kind of functions has $\mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3$ as domain, i.e. they receive a n-th order tensor as argument. They can be further subdivided according to their output,

1. Scalar-valued: $f : \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3 \mapsto \mathbb{R}$. Example: a quadratic form, $q(\boldsymbol{A}) = \boldsymbol{A} : \mathbb{C} : \boldsymbol{A} \in \mathbb{R}$.

2. Vector-valued: $f : \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3 \mapsto \mathbb{R}^3$. Example: axial vector of a skew-symmetric tensor, $\boldsymbol{\varphi}(\boldsymbol{W}) = \boldsymbol{\varepsilon} : \boldsymbol{W} \in \mathbb{R}^3$, where $\boldsymbol{\varepsilon}$ is the third-order permutation tensor.

3. Tensor-valued: $f : \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3 \mapsto \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3$. Example: the tensor product with another tensor, $\mathbb{C}(\boldsymbol{A}) = \boldsymbol{A} \otimes \boldsymbol{B} \in \mathbb{R}^3 \otimes \mathbb{R}^3$.

2.1.4. *Space of tensors*

Functions are mappings between vector spaces. In our context, we shall define a generic notation for a tensor space, since most of our functions will be generalized to mappings from and to tensor spaces. We, thus, will define the space of p-th order tensors in three-dimensional euclidean space as

$$\mathcal{T}^p = \underbrace{\mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3}_{p \text{ indices}}. \tag{61}$$

2.1.5. *Common analytic functions using tensor arguments*

Analytic functions are functions that can locally (close to a point in the domain) be represented by a convergent power series of its argument. Examples of such functions are polynomials, n-th roots, exponentiation, logarithm, sine, and cosine. The aforementioned analytic functions can be represented by Taylor series, for instance. Thus, such functions can be extended to tensor arguments using the same rationale—plugging the tensor argument directly into the power series [5]. To illustrate this concept, we present the *exponential of a tensor*,

$$\exp(\boldsymbol{A}) = \sum_{n=0}^{\infty} \frac{1}{n!}\boldsymbol{A}^n = \boldsymbol{I} + \boldsymbol{A} + \frac{1}{2!}\boldsymbol{A}^2 + \frac{1}{3!}\boldsymbol{A}^3 + \dots . \tag{62}$$

It becomes clear that analytic functions of tensors are mappings $f : \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3 \mapsto \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3$. We refer to the exponential of a tensor for it naturally appears as solution to ordinary differential equations, such as the time-evolution of stress in viscoelastic materials.

As the natural logarithm is the inverse function, $f^{-1}\big(f(\boldsymbol{A})\big) = \boldsymbol{A}$, of the exponential function, we shall stand its Taylor series for completeness,

$$\log(\boldsymbol{A} + \boldsymbol{I}) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}\boldsymbol{A}^n = \boldsymbol{A} - \frac{1}{2}\boldsymbol{A}^2 + \frac{1}{3}\boldsymbol{A}^3 + \dots . \tag{63}$$

Challenge 2.1

Prove that the rotation tensor $\boldsymbol{R}(\boldsymbol{\varphi})$, given by the Euler-Rodrigues formula (see Eq. (41)), is equivalent to $\boldsymbol{R}(\boldsymbol{\varphi}) = \exp\left(\boldsymbol{W}(\boldsymbol{\varphi})\right) = \exp(\boldsymbol{\varepsilon}\cdot\boldsymbol{\varphi})$. Conversely, show that $\boldsymbol{\varphi} = \operatorname{axl}\left(\log(\boldsymbol{R})\right)$.

2.2. Derivatives

Once functions of tensors have been introduced, it is straightforward to extend calculus to these objects. We shall launch this endeavor towards derivatives first, for their relative simplicity and major application thereafter in integration. We will walk step by step through differentiation techniques to join tensor differentiation to undergraduate-level derivatives, as seamless as possible. The most important advice is to differentiate a tensor by taking derivatives of each one of its components (which are themselves scalars); hence, common differentiation techniques will be readily available and applicable.

2.2.1. Recollecting the product and the chain rules

Suppose that a function $h(t) : \mathbb{R} \mapsto \mathbb{R}$ is equal to the multiplication of two other functions, $f(t) : \mathbb{R} \mapsto \mathbb{R}$ and $g(t) : \mathbb{R} \mapsto \mathbb{R}$, such that $h(t) = f(t)\, g(t)$. Then, the derivative of h with respect to t is given by the *product rule*,

$$\frac{\partial h}{\partial t} = \frac{\partial f}{\partial t} g(t) + f(t)\frac{\partial g}{\partial t}. \tag{64}$$

Conceive a function $h(t) : \mathbb{R} \mapsto \mathbb{R}$ that is equal to the composition of two other functions, $f(t) : \mathbb{R} \mapsto \mathbb{R}$ and $g(t) : \mathbb{R} \mapsto \mathbb{R}$, such that $h(t) = f\left(g(t)\right)$. Then, the derivative of h with respect to t is given by the *chain rule*,

$$\frac{\partial h}{\partial t} = \frac{\partial f}{\partial g}\frac{\partial g}{\partial t}. \tag{65}$$

Imagine that there is a multi-variable scalar-valued function $g(t) : \mathbb{R}^m \mapsto \mathbb{R},\ m \in \mathbb{Z}^+$. Suppose, as well, that there are m scalar-valued functions $f_i(t) : \mathbb{R} \mapsto \mathbb{R}\ \ i \in \{1, 2, \ldots, m\}$. Then, g can be composed with the f_i functions to generate $h(t) = g\left(f_1(t), f_2(t), \ldots, f_m(t)\right)$. The derivative of h with respect to t is readily given by the chain rule,

$$\frac{\partial h}{\partial t} = \frac{\partial g}{\partial f_1}\frac{\partial f_1}{\partial t} + \frac{\partial g}{\partial f_2}\frac{\partial f_2}{\partial t} + \cdots + \frac{\partial g}{\partial f_m}\frac{\partial f_m}{\partial t}. \tag{66}$$

2.2.2. Derivative of a tensor-valued function of a single variable

Let a tensor-valued function of a single variable, $f(t) : \mathbb{R} \mapsto \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3$, such that

$$\boldsymbol{A} = f(t) = A_{ij\ldots s}(t)\, \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_s. \tag{67}$$

Then, the derivative of this tensor with respect to t is given component by component,

$$\dot{\boldsymbol{A}} = \frac{\partial \boldsymbol{A}}{\partial t} = \frac{\partial A_{ij\ldots s}}{\partial t}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_s = \dot{A}_{ij\ldots s}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_s; \tag{68}$$

where $\dot{A}$ is an alternative (and simpler) notation for differentiation of a function that depends on a single variable.

Examples of this operation in continuum mechanics can be found in the evaluation of rates (time derivatives), such as the rates of: displacement, to calculate velocity; velocity, to compute acceleration; strain, to compute the evolution of strain with time.

Exercise 2.1

Use the derivative definition (Newton quotient),

$$\left.\frac{\partial f}{\partial t}\right|_t = \lim_{\epsilon \to 0} \frac{f(t+\epsilon) - f(t)}{\epsilon}, \tag{69}$$

to prove Eq. (68) using Einstein notation.

2.2.3. Derivative of a scalar-valued function of tensor argument

Let a scalar-valued function of tensor argument, $f : \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3 \mapsto \mathbb{R}$. This function can be rewritten as a multi-variable function of the individual components of the tensor arguments, without loss of generality,

$$\boldsymbol{A} = A_{ij\ldots s}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_s \implies f(\boldsymbol{A}) = f(A_{11\ldots 1}, A_{11\ldots 2}, A_{11\ldots 3}, \ldots, A_{12\ldots 1}, \ldots, A_{33\ldots 3}). \tag{70}$$

Thus, using Eq. (66) and, then, organizing the multiple derivatives into a tensor format, the derivative of f with respect to $\boldsymbol{A}$ becomes

$$\frac{\partial f}{\partial \boldsymbol{A}} = \frac{\partial f}{\partial A_{ij\ldots s}}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_s \in \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3. \tag{71}$$

Note that the derivative of a scalar-valued function of tensor argument with respect to its argument becomes a tensor, hence, this derivative is a mapping from $\mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3$ to $\mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3$. Applications of this derivative can be found in the linearization (Taylor series truncated at the linear term) of energy and power functions, and in the evaluation of stresses. One particular worth-noting example is the derivative of the determinant of a second-order tensor,

$$\frac{\partial}{\partial \boldsymbol{A}}\det(\boldsymbol{A}) = \det(\boldsymbol{A})\,\boldsymbol{A}^{-\mathrm{T}}. \tag{72}$$

Exercise 2.2

Evaluate $\frac{\partial}{\partial \boldsymbol{A}}\text{tr}\left(\boldsymbol{A}\right)$, where $\boldsymbol{A}$ is a second-order tensor.

Exercise 2.3

The euclidean norm can be written as the square root of the simple contraction between the vector and itself, $\|\boldsymbol{v}\|_2 = \sqrt{\boldsymbol{v}\cdot\boldsymbol{v}}$. Use this fact to compute $\frac{\partial}{\partial \boldsymbol{v}}\|\boldsymbol{v}\|_2$. Show that this result is a vector (first-order tensor).

Challenge 2.2

Use the Cayley-Hamilton theorem (see Eq. (55)) to demonstrate Eq. (72).

2.2.4. Derivative of a tensor-valued function of tensor argument

Suppose that there is a tensor-valued function of tensor argument $f:\mathbb{R}^3\otimes\mathbb{R}^3\otimes\cdots\otimes\mathbb{R}^3\mapsto\mathbb{R}^3\otimes\mathbb{R}^3\otimes\cdots\otimes\mathbb{R}^3$. Each component of the image tensor (the output tensor) can be written as a multi-variable scalar-valued function of the components of the domain tensor (the input tensor),

$$\begin{aligned}\boldsymbol{A} = A_{ij\ldots s}\boldsymbol{e}_i\otimes\boldsymbol{e}_j\otimes\cdots\otimes\boldsymbol{e}_s,\ \boldsymbol{B} = B_{\alpha\beta\ldots\xi}\boldsymbol{e}_\alpha\otimes\boldsymbol{e}_\beta\otimes\cdots\otimes\boldsymbol{e}_\xi,\\ \boldsymbol{A} = f\left(\boldsymbol{B}\right) = A_{ij\ldots s}\left(B_{11\ldots1}, B_{11\ldots2}, B_{11\ldots3},\ldots, B_{12\ldots1},\ldots, B_{33\ldots3}\right)\boldsymbol{e}_i\otimes\boldsymbol{e}_j\otimes\cdots\otimes\boldsymbol{e}_s.\end{aligned} \tag{73}$$

Consequently, a generic component of the image tensor can be differentiated with respect to the components of the domain tensor, following the same procedures of Subsubsec. 2.2.3, to retrieve a tensor,

$$\frac{\partial A_{ij\ldots s}}{\partial\boldsymbol{B}} = \frac{\partial A_{ij\ldots s}}{\partial B_{\alpha\beta\ldots\xi}}\boldsymbol{e}_\alpha\otimes\boldsymbol{e}_\beta\otimes\cdots\otimes\boldsymbol{e}_\xi. \tag{74}$$

Equation (74) can be used to evaluate the derivative of f with respect to the domain tensor,

$$\frac{\partial\boldsymbol{A}}{\partial\boldsymbol{B}} = \boldsymbol{e}_i\otimes\boldsymbol{e}_j\otimes\cdots\otimes\boldsymbol{e}_s\otimes\frac{\partial A_{ij\ldots s}}{\partial\boldsymbol{B}} = \frac{\partial A_{ij\ldots s}}{\partial B_{\alpha\beta\ldots\xi}}\boldsymbol{e}_i\otimes\boldsymbol{e}_j\otimes\cdots\otimes\boldsymbol{e}_s\otimes\boldsymbol{e}_\alpha\otimes\boldsymbol{e}_\beta\otimes\cdots\otimes\boldsymbol{e}_\xi. \tag{75}$$

Note that the basis vectors of the domain tensor were positioned to the *right*, after the vectors of the image tensor, this is a **convention**—known as *right-differentiation*. Right-differentiation will be the convention of the whole of this text.

Observe that the result of the derivative of a n-th order tensor with respect to a m-th order tensor is a $(n+m)$-th order tensor. Physically (and mathematically), it means that the sensitivity of all components of the image tensor is evaluated with respect to all components of the domain tensor; thus, ensembling $3^n3^m = 3^{n+m}$ combinations, that is exactly the number of components in a $(n+m)$-th order tensor.

Exercise 2.4

Use Eq. (64) and Eq. (75) to show that, given a vector $\boldsymbol{v}$,

$$\frac{\partial}{\partial \boldsymbol{v}}\left[\frac{\boldsymbol{v}}{\|\boldsymbol{v}\|_2}\right] = \frac{1}{\|\boldsymbol{v}\|_2}\boldsymbol{I} - \frac{1}{\|\boldsymbol{v}\|_2^3}\boldsymbol{v}\otimes\boldsymbol{v}. \tag{76}$$

Note: this exercise is particularly useful for numerical implementation of infinitesimal plastic constitutive models.

2.2.5. Chain rule between tensor-valued functions

This subsubsection is dedicated to show, through one example, how the double contraction naturally appears as a consequence of the chain rule. Additionally, the presented example will be used later, especially in Part II.

Consider a second-order tensor $\boldsymbol{P}$ that is function of another second-order tensor, $\boldsymbol{F}$. Tensor $\boldsymbol{F}$ can be a function of a single variable, time for instance. Thus, the tensor $\boldsymbol{P}$ is also a function of time, $\boldsymbol{P} = \boldsymbol{P}\left(\boldsymbol{F}(t)\right)$. It is perfectly plausible to require the derivative of $\boldsymbol{P}$ with respect to t to evaluate its rate of change in time. This rate can be evaluated component by component,

$$\dot{\boldsymbol{P}} = \frac{\partial \boldsymbol{P}}{\partial t} = \frac{\partial}{\partial t}P_{ij}\left(F_{11}, F_{12}, F_{13}, F_{21}, F_{22}, F_{23}, F_{31}, F_{32}, F_{33}\right)\boldsymbol{e}_i\otimes\boldsymbol{e}_j. \tag{77}$$

Using Eq. (66), Equation (77) can be rewritten as

$$\begin{aligned} &\dot{\boldsymbol{P}} = \left[\frac{\partial P_{ij}}{\partial F_{11}}\dot{F}_{11} + \frac{\partial P_{ij}}{\partial F_{12}}\dot{F}_{12} + \frac{\partial P_{ij}}{\partial F_{13}}\dot{F}_{13} + \cdots + \frac{\partial P_{ij}}{\partial F_{33}}\dot{F}_{33}\right]\boldsymbol{e}_i\otimes\boldsymbol{e}_j \\ \xrightarrow[\text{notation}]{\text{Einstein}}\ &\dot{\boldsymbol{P}} = \frac{\partial P_{ij}}{\partial F_{kl}}\dot{F}_{kl}\boldsymbol{e}_i\otimes\boldsymbol{e}_j = \left(\frac{\partial P_{ij}}{\partial F_{kl}}\boldsymbol{e}_i\otimes\boldsymbol{e}_j\otimes\boldsymbol{e}_k\otimes\boldsymbol{e}_l\right) : \left(\dot{F}_{mn}\boldsymbol{e}_m\otimes\boldsymbol{e}_n\right) = \frac{\partial \boldsymbol{P}}{\partial \boldsymbol{F}} : \dot{\boldsymbol{F}}; \end{aligned} \tag{78}$$

where $\frac{\partial \boldsymbol{P}}{\partial \boldsymbol{F}}$ is a fourth-order tensor. Note that the basis vector $\boldsymbol{e}_k$ and $\boldsymbol{e}_l$ naturally appeared to the right as a consequence of our choice of positioning of the chain rule.

Exercise 2.5

Let a second-order tensor $\boldsymbol{A}$ be a function of another second-order tensor, $\boldsymbol{B}$, which, in turn, is a function of a vector $\boldsymbol{v}$. Show, using differentiation by components and Einstein notation, that

$$\frac{\partial \boldsymbol{A}}{\partial \boldsymbol{v}} = \frac{\partial \boldsymbol{A}}{\partial \boldsymbol{B}} : \frac{\partial \boldsymbol{B}}{\partial \boldsymbol{v}}. \tag{79}$$

2.2.6. Directional derivative

Consider a tensor-valued function of tensor argument $f : \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3 \mapsto \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes \cdots \otimes \mathbb{R}^3$. Sometimes, we do not want to evaluate the *complete* sensitivity of a function in a point; rather to evaluate its sensitivity in a *direction* from a point. The latter is known as a *directional derivative* and, much like a conventional derivative, it is defined by a limit,

$$D_{\boldsymbol{B}} f|_{\boldsymbol{A}} = \lim_{\epsilon \to 0} \left[\frac{f(\boldsymbol{A} + \epsilon \boldsymbol{B}) - f(\boldsymbol{A})}{\epsilon} \right]; \tag{80}$$

which reads as: *the directional derivative of f in the direction of $\boldsymbol{B}$ evaluated at $\boldsymbol{A}$.*

Observe that, if the image of f is a n-th order tensor, the directional derivative of f in the direction of $\boldsymbol{B}$, a m-th order tensor, is still a n-th order tensor. Thus, in contrast to the conventional partial derivative, the order of the tensor resulting of the directional derivative is equal to the order of the image tensor of the differentiated function itself.

Remark 2.1

If f is continuous and differentiable, and $\boldsymbol{B}$ is unitary (i.e. $\|\boldsymbol{B}\|_2 = 1$), the directional derivative of f in the direction of $\boldsymbol{B}$ can be evaluated by

$$D_{\boldsymbol{B}} f = \frac{\partial f}{\partial \boldsymbol{B}} \odot \boldsymbol{B}, \tag{81}$$

in which $\odot$ is a suitable contraction operation that removes as many indices as added by the partial derivatives. For the particular case of second-order tensors,

$$D_{\boldsymbol{B}} f = \frac{\partial f}{\partial \boldsymbol{B}} : \boldsymbol{B}, \quad \boldsymbol{B} \in \mathbb{R}^3 \otimes \mathbb{R}^3. \tag{82}$$

The geometric interpretation of Eq. (81) is the *projection of the partial derivative with respect to the argument onto the unitary direction $\boldsymbol{B}$.*

2.2.7. Taylor series of a tensor-valued function of tensor argument

A Taylor series of a tensor-valued function of tensor argument can be constructed using the definition of a Taylor, with the difference that the derivatives are with respect to a tensor argument. For this reason, there must be appropriate contraction operations to contract the indices gained by differentiation with the indices of the step from the approximation point. We first define this *appropriate contraction operation*, named the n-th contraction, as

$$\boldsymbol{A} \odot^n \boldsymbol{B} = A_{ij\ldots s} \underbrace{\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_s}_{m+n \text{ indices}} \odot^n B_{\alpha\beta\ldots\xi} \underbrace{\boldsymbol{e}_\alpha \otimes \boldsymbol{e}_\beta \otimes \cdots \otimes \boldsymbol{e}_\xi}_{n \text{ indices}}$$
$$= A_{ij\ldots opq\ldots s} B_{pq\ldots s} \underbrace{\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_o}_{m \text{ indices}}. \tag{83}$$

Thereafter, the Taylor series can be constructed using the definition for a function $f : \mathcal{T}^m \mapsto \mathcal{T}^n$, where $\mathcal{T}^p$ is the space of p-th order tensors in three-dimensional space (see Eq. (61)),

$$\begin{aligned} f(\boldsymbol{X} + \Delta\boldsymbol{X}) &= f(\boldsymbol{X}) + \left.\frac{\partial f}{\partial \boldsymbol{X}}\right|_{\boldsymbol{X}} \cdot \Delta\boldsymbol{X} + \frac{1}{2}\left(\left.\frac{\partial^2 f}{\partial \boldsymbol{X}^2}\right|_{\boldsymbol{X}} \cdot \Delta\boldsymbol{X}\right) \cdot \Delta\boldsymbol{X} \\ &\quad + \frac{1}{6}\left(\left(\left.\frac{\partial^3 f}{\partial \boldsymbol{X}^3}\right|_{\boldsymbol{X}} \cdot \Delta\boldsymbol{X}\right) \cdot \Delta\boldsymbol{X}\right) \cdot \Delta\boldsymbol{X} + \ldots \\ &= f(\boldsymbol{X}) + \left.\frac{\partial f}{\partial \boldsymbol{X}}\right|_{\boldsymbol{X}} \cdot \Delta\boldsymbol{X} + \frac{1}{2}\left.\frac{\partial^2 f}{\partial \boldsymbol{X}^2}\right|_{\boldsymbol{X}} : (\Delta\boldsymbol{X} \otimes \Delta\boldsymbol{X}) \\ &\quad + \frac{1}{6}\left.\frac{\partial^3 f}{\partial \boldsymbol{X}^3}\right|_{\boldsymbol{X}} \vdots\, (\Delta\boldsymbol{X} \otimes \Delta\boldsymbol{X} \otimes \Delta\boldsymbol{X}) + \ldots; \end{aligned} \tag{84}$$

or, in compact notation,

$$f(\boldsymbol{X} + \Delta\boldsymbol{X}) = f(\boldsymbol{X}) + \sum_{k=1}^{\infty}\left[\frac{1}{k!}\left.\frac{\partial^k f}{\partial \boldsymbol{X}^k}\right|_{\boldsymbol{X}} \odot^k \underbrace{(\Delta\boldsymbol{X} \otimes \Delta\boldsymbol{X} \otimes \cdots \otimes \Delta\boldsymbol{X})}_{k \text{ times}}\right]. \tag{85}$$

Note that, once again, the right-differentiation convention is fundamental to securing proper contraction.

2.3. Integration

Integration of tensors is, in actuality, simpler than taking their derivatives. We restrain ourselves, for now, to show the following identity of the integration of a tensor in a measure of space (volume, area, or length),

$$\int \boldsymbol{A}\, \mathrm{d}M = \int A_{ij\ldots s}\, \mathrm{d}M \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_s. \tag{86}$$

Further details will be given as they are needed for physical derivations.

3. Kinematics

Kinematics is the observation or study of motion in a body or an ensemble of bodies. When someone is introduced to the term *motion*, the first thought is usually rigid-body motions, such as translations, rotations, and combinations of them. In continuum mechanics, we are concerned with a richer type of motion, that of *deformable bodies*. During such motion, a *portion of material* not only translates and rotates relative to a reference system of coordinates, but also each *point* in this portion moves relative to other *points* in it. Visibly, this translates to change in shape and or volume of the said portion of material.

3.1. Main hypothesis of the continuum

In continuum mechanics, in contrast to quantum mechanics for instance, matter is *infinitely divisible*; hence the *continuum* predicate. This means that there is no primordial unit of matter, there is not an *atom*—the smallest possible constituent of matter. In contrast, any body (solid, liquid, or gas) is a collection of an uncountable and infinite number of *points*. Each point is a dimensionless particle whose motion is governed by the laws of Newton.

The objective of *modeling* matter as a continuum lies behind calculus. To calculate derivatives, limits must be taken, and they must exist. Thus, any function must be evaluated as close to a point as an *infinitesimal*. Derivatives are important to track how a point moves relative to its neighbor points, and, therefore, derivatives will be crucial to define strain.

We will restrain ourselves to a continuum whose points are particles submitted to translation only; thus, they do not have any underlying structure that interacts with localized torques to, consequently, respond with rotations. Such continuum is said to be *non-polar* and the primary kinematic field is displacement or velocity. However, there are continuum theories that have rotation fields as well as displacement, such as micropolar theory [3, 6].

3.2. Motion, mapping and configuration

A body moves in space, thus, each one of the infinite particles that compose it move in space as well. If we took snapshots in time, we would observe different positions for each particle relative to a system of coordinates. Each one of these snapshots is called a *configuration*, which, mathematically, contains all the necessary information to point out the position and state of each and every particle of the continuum.

We can *uniquely* identify each particle from its *position vector*, $\boldsymbol{X}$, in the *reference configuration* (or *material configuration*), i.e. at time $t = 0$, in other words, at the time the motion began to be observed. Conversely, we can identify the same particle by its position in the *deformed configuration* $\boldsymbol{x}$, also known as *current configuration* or *spatial configuration*, which presents the body at the present moment. See Fig. 1 to verify that the same particle is featured in the reference and deformed configurations, observe the illustration of the position vectors $\boldsymbol{X}$ and $\boldsymbol{x}$. The position vector in the reference configuration, $\boldsymbol{X}$, is usually called the *material position vector*; conversely, $\boldsymbol{x}$ is denoted as the *spatial position vector*.

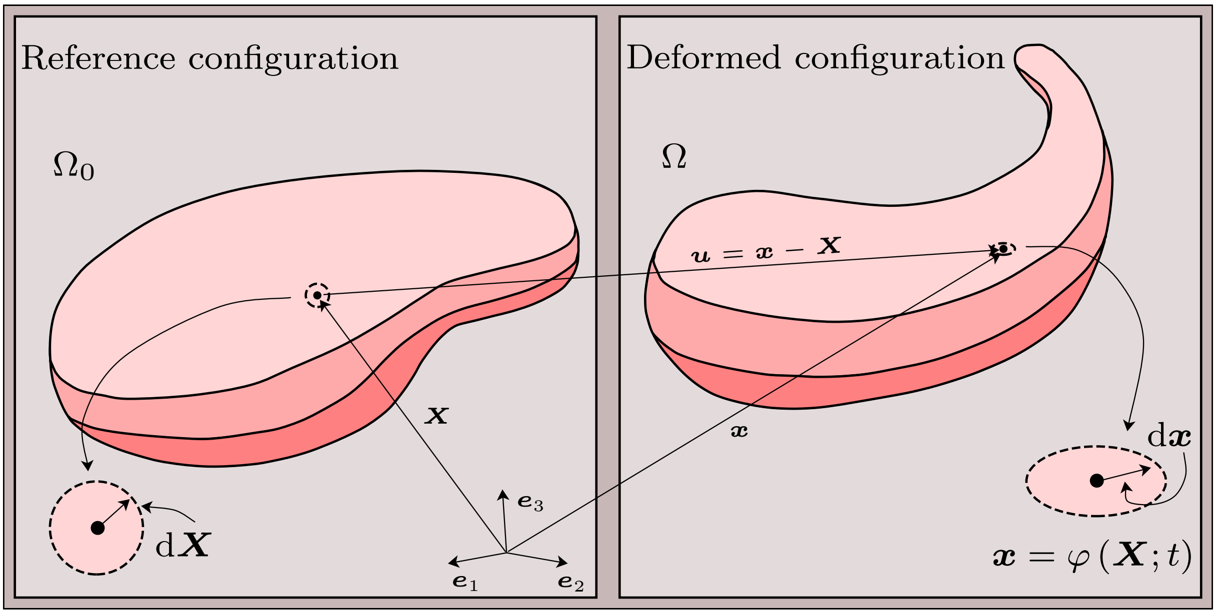

Figure 1: Kinematics of a generic body. The right-hand-side frame features the reference configuration, whose domain is Ω_0, and a particle with a vicinity. The particle in the reference configuration is identified to the system of coordinates by the material position vector $\boldsymbol{X}$. The left-hand-side shows the deformed (current) configuration, where the same particle is referenced by the spatial position vector $\boldsymbol{x}$. The vicinity of the particle is zoomed in and shown in both configuration, a change of shape of this vicinity is presented. Additionally, an infinitesimal vector contained within the vicinity is shown in both configuration, there is a clear change of orientation.

As the particle is the same in all configurations, there must be a function that maps one to another. This function is called *mapping*, and it captures the motion of any particle at any time,

$$\boldsymbol{x} = \varphi(\boldsymbol{X}, t) \rightarrow \varphi : \mathbb{R}^3 \times \mathbb{R} \mapsto \mathbb{R}^3. \tag{87}$$

The mapping function φ must be a bijection for the following reasons: (i) the particle is the same across configurations; (ii) the configurations live in the same vector space; (iii) two particles cannot occupy the same point in space at the same time.

Remark 3.1

The reference configuration, unintuitively, is not the **frozen** state of deformation (motion) in the first possible time the deformation process began to be observed and recorded. Rather, the reference configuration possesses all the knowledge of the deformation at any time, with the restriction that it is contained by each particle in the position they occupied in the domain of the reference configuration. This is achieved by the mapping we discussed before. The reference configuration is a mathematical construction to perform integration without updating the integration domain, since, in finite strains, the spatial configuration domain, Ω, changes significantly. One has to keep in mind that the physical phenomenon always happens in the spatial configuration.

Physically, a bijective mapping between configurations translates to the impossibility of a three-dimensional chunk of matter ever being squashed into a plane, a line, or any other topological object with less dimensions than 3.

We can write the Taylor series (see Eq. (85)) of the mapping function using an infinitesimal direction in the material configuration,

$$\begin{aligned}\boldsymbol{x}\left(\boldsymbol{X}+\mathrm{d}\boldsymbol{X},t\right) &= \varphi\left(\boldsymbol{X},t\right)+\left.\frac{\partial\varphi}{\partial\boldsymbol{X}}\right|_{\boldsymbol{X},t}\mathrm{d}\boldsymbol{X}+\frac{1}{2}\left.\frac{\partial^2\varphi}{\partial\boldsymbol{X}^2}\right|_{\boldsymbol{X},t}:\left(\mathrm{d}\boldsymbol{X}\otimes\mathrm{d}\boldsymbol{X}\right)+\ldots\\ \boldsymbol{x}\left(\boldsymbol{X}+\mathrm{d}\boldsymbol{X},t\right) &= \varphi\left(\boldsymbol{X},t\right)+\mathrm{d}\boldsymbol{x}+\mathrm{d}\boldsymbol{x}^2+\ldots,\end{aligned} \tag{88}$$

where $\mathrm{d}\boldsymbol{x}^k$ is the k-th order directional derivative of the mapping function in the direction of the infinitesimal material direction $\mathrm{d}\boldsymbol{X}$. If we neglect the higher-order directional derivatives due to the decreasing norm of the multiple tensor products of $\mathrm{d}\boldsymbol{X}$ (keep in mind that this directional has infinitesimal length), the mapping will be locally approximated by a linear transformation,

$$\boldsymbol{x}\left(\boldsymbol{X}+\mathrm{d}\boldsymbol{X},t\right)\approx\boldsymbol{x}\left(\boldsymbol{X},t\right)+\mathrm{d}\boldsymbol{x}=\boldsymbol{x}\left(\boldsymbol{X},t\right)+\left.\frac{\partial\varphi}{\partial\boldsymbol{X}}\right|_{\boldsymbol{X},t}\mathrm{d}\boldsymbol{X}=\boldsymbol{x}\left(\boldsymbol{X},t\right)+\boldsymbol{F}\mathrm{d}\boldsymbol{X}, \tag{89}$$

where $\boldsymbol{F}$ is called the *deformation gradient*, a second-order tensor and one of the most important tensors in continuum mechanics. Observe that the deformation gradient, as the kernel of the linear transformation, is responsible to projecting vectors from the material configuration onto the *tangent space* of the deformed (spatial) configuration. Thus, the deformation gradient tensor will appear in all derivatives relating both configurations (due to the chain rule).

Remark 3.2

In a mathematical sense (especially in differential geometry), the deformation gradient would be denoted as the *jacobian* of the transformation of vectors from the reference to the deformed configurations. We, however, will not use this terminology, because the name *jacobian*, in continuum mechanics, is historically reserved to denote the determinant of the deformation gradient.

With the help of Fig. 1, it is possible to see that the displacement vector of the particle from the reference configuration to the deformed configuration is given by

$$\begin{aligned}\boldsymbol{u}\left(\boldsymbol{X},t\right) &= \boldsymbol{x}\left(\boldsymbol{X},t\right)-\boldsymbol{X}\\ =\boldsymbol{u}\left(\boldsymbol{x},t\right) &= \boldsymbol{x}\left(t\right)-\boldsymbol{X}\left(\boldsymbol{x}\right)=\boldsymbol{x}\left(t\right)-\varphi^{-1}\left(\boldsymbol{x},t\right).\end{aligned} \tag{90}$$

Thus, it is possible to relate the deformation gradient tensor with the displacement field through the displacement function of the material position vector,

$$\frac{\partial \boldsymbol{u}}{\partial \boldsymbol{X}} = \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{X}} - \frac{\partial \boldsymbol{X}}{\partial \boldsymbol{X}} = \boldsymbol{F} - \boldsymbol{I} \implies \boldsymbol{F} = \boldsymbol{I} + \frac{\partial \boldsymbol{u}}{\partial \boldsymbol{X}}. \tag{91}$$

Inspecting Eq. (90), it is possible to differentiate the displacement field with respect to the material position vector using the chain rule, since the displacement field can be a function of the spatial position vector,

$$\frac{\partial \boldsymbol{u}}{\partial \boldsymbol{X}} = \frac{\partial \boldsymbol{u}}{\partial \boldsymbol{x}}\frac{\partial \boldsymbol{x}}{\partial \boldsymbol{X}} = \frac{\partial \boldsymbol{u}}{\partial \boldsymbol{x}}\boldsymbol{F}. \tag{92}$$

The partial derivative of a field with respect to a position vector is denoted as the *gradient of the field*. In continuum mechanics, there are different configurations, thus, there are multiple position vectors and, for this reason, distinct possible evaluations of the gradient operator.

We note, for instance, two different gradient operations in Eq. (92)—one with respect to the material vector and another with respect to the spatial position vector. Consequently, we can define two gradient operators,

$$\begin{aligned} \text{material gradient: } \operatorname{Grad}(\cdot) &= \frac{\partial \cdot}{\partial \boldsymbol{X}}, \\ \text{spatial gradient: } \operatorname{grad}(\cdot) &= \frac{\partial \cdot}{\partial \boldsymbol{x}}. \end{aligned} \tag{93}$$

For the particular case of evaluating the gradient of vector fields, such as displacement in Eq. (92), the two gradients relate to each other by the following rule,

$$\operatorname{Grad}(\boldsymbol{v}) = \operatorname{grad}(\boldsymbol{v})\,\boldsymbol{F}, \tag{94}$$

$$\operatorname{grad}(\boldsymbol{v}) = \operatorname{Grad}(\boldsymbol{v})\,\boldsymbol{F}^{-1}. \tag{95}$$

We refer to Eq. (94) as the *pull-back* operation of the spatial gradient of a vector field to the material gradient of such a field. Whereas, in Eq. (95), we call it the *push-forward* of the material gradient of a vector field to the gradient of this field in the spatial configuration. Such terminology is particularly useful for condensing the meaning of transforming operations from one configuration to the other. In summary: **push-forward** transforms from the material to the spatial configurations (since the spatial configuration is forward in time); **pull-back** transforms from the spatial to the material configurations.

In many derivations ahead, the velocity vector field will be necessary, such as for evaluating derivatives in the spatial configuration and power. The velocity vector is by definition the time derivative of the displacement field, hence using Eq. (90),

$$\boldsymbol{v} = \frac{\mathrm{d}\boldsymbol{u}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left[\boldsymbol{x} - \boldsymbol{X}\right] = \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t}, \tag{96}$$

which is the time derivative of the spatial position vector since the material position vector is fixed in time. Furthermore, the velocity vector can be related to the time derivative of the deformation gradient tensor using Eq. (91),

$$\frac{\mathrm{d}\boldsymbol{F}}{\mathrm{d}t} = \dot{\boldsymbol{F}} = \frac{\mathrm{d}}{\mathrm{d}t}\left[\boldsymbol{I} + \mathrm{Grad}\left(\boldsymbol{u}\right)\right] = \mathrm{Grad}\left(\dot{\boldsymbol{u}}\right) = \mathrm{Grad}\left(\boldsymbol{v}\right). \tag{97}$$

We stress that the time derivative of the gradient of the displacement has been directly applied to the displacement field only because that gradient is defined upon the reference configuration, whose position vector is fixed in time.

Remark 3.3

Push-forward and pull-back are common operations in differential geometry, and they are used in the same context, mapping in between manifolds. Such operations will be seen all over continuum mechanics, but also in Part II, where they will play a crucial role in numerical integration of objects within the finite element method.

3.3. Gradient and divergence in material and spatial configurations

Suppose that there is a function $f : \mathcal{T}^n \mapsto \mathcal{T}^m$ (see Eq. (61) for the definition of $\mathcal{T}^p$) mapping one tensor space to another. We can define the material gradient of f as the tensor product to the right of the partial derivative operator with respect to the material position vector,

$$\mathrm{Grad}\left(f\right) = f \otimes \frac{\partial}{\partial \boldsymbol{X}} = \frac{\partial f_{ij\ldots s}}{\partial X_\alpha}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_s \otimes \boldsymbol{e}_\alpha, \tag{98}$$

where $f_{ij\ldots s}$ are the components of the image tensor of function f. Observe, once more, that the gradient operator added one index to the tensor and that the corresponding basis vector was multiplied to the right.

The divergence operator is also given by the partial derivative with respect to the position vector, but the algebraic operation is a simple contraction, rather than the tensor product. Due to the partial derivative with respect to the position vector, there is also a distinction between material and spatial divergences. Let us start with the material divergence,

$$\begin{aligned}\mathrm{Div}\left(f\right) = f \cdot \frac{\partial}{\partial \boldsymbol{X}} &= \left(f_{ij\ldots rs}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_r \otimes \boldsymbol{e}_s\right) \cdot \left(\frac{\partial}{\partial X_\alpha}\boldsymbol{e}_\alpha\right) \\ &= \frac{\partial f_{ij\ldots rs}}{\partial X_s}\boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_r. \end{aligned} \tag{99}$$

Finally, it is straightforward to show that the spatial gradient is, thus, given by

$$\operatorname{grad}(f) = f \otimes \frac{\partial}{\partial \boldsymbol{x}} = \frac{\partial f_{ij\ldots r}}{\partial x_\alpha} \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_s \otimes \boldsymbol{e}_\alpha; \tag{100}$$

and the spatial divergence as

$$\operatorname{div}(f) = f \cdot \frac{\partial}{\partial \boldsymbol{x}} = \frac{\partial f_{ij\ldots rs}}{\partial x_s} \boldsymbol{e}_i \otimes \boldsymbol{e}_j \otimes \cdots \otimes \boldsymbol{e}_r. \tag{101}$$

The gradient and divergence operators possess a profound physical meaning. The gradient is a direction that points to the path where the function instantly increases most—to illustrate this, picture yourself on a flat surface, you can walk to any direction but only on said surface, suppose that there is a function of the two coordinates of such surface; then, if you evaluate the gradient of this function with respect to the coordinates of the surface and you walk towards this direction, this way the function will increase in value the most. Divergence, in contrast, tells if a vector field is conservative in an infinitesimal vicinity to a point; i.e. if the divergence is zero, for all vectors that *go into the vicinity*, there are vectors that *leave the vicinity behind*.

Continuum mechanics uses gradients to express *changes*, such as shape, stress, and flux; whereas divergence is used to express conservation laws, such as conservation of mass, energy, and momentum. Many natural phenomena, such as heat transfer, sound propagation and the deformation of bodies, are modeled using the conservation of a flux generated by the gradient of a potential. Hence, the divergence of a gradient emerges, which is called in mathematics the Laplace operator and, eventually, leads to Laplace partial differential equations. We will not treat Laplace operators directly, since we will not be able to directly apply the divergence operator to the result of the gradient of a potential. The inapplicability of the direct use of Laplace operator naturally arises in anisotropic and nonlinear problems.

Exercise 3.1

i) Develop the pull-back and push-forward for the gradient operator, between Eq. (98) and Eq. (100), and for the divergence operator, between Eq. (99) and Eq. (101).

ii) Show that the gradient of a scalar field $f : \mathbb{R}^3 \mapsto \mathbb{R}$ is a vector field.

iii) Show that the divergence of a vector field $f : \mathbb{R}^3 \mapsto \mathbb{R}^3$ is a scalar field; then, show that the divergence of second-order tensor field $f : \mathbb{R}^3 \mapsto \mathcal{T}^2$ is a vector field.

iv) Given the displacement field $\boldsymbol{u}$, show that the trace of the deformation gradient is equal to the material divergence of $\boldsymbol{u}$ plus three, ie. $\operatorname{tr}(\boldsymbol{F}) = \operatorname{Div}(\boldsymbol{u}) + 3$.

Exercise 3.2

Show that, for any first-order tensor $\boldsymbol{a}$, the following identity holds,

$$\begin{aligned} \mathrm{div}\left(\boldsymbol{a}\right) &= \mathrm{tr}\left(\mathrm{grad}\left(\boldsymbol{a}\right)\right), \\ \mathrm{Div}\left(\boldsymbol{a}\right) &= \mathrm{tr}\left(\mathrm{Grad}\left(\boldsymbol{a}\right)\right). \end{aligned} \tag{102}$$

Challenge 3.1

Pick three generic infinitesimal vectors in the reference configuration, $\mathrm{d}\boldsymbol{X}$, $\mathrm{d}\boldsymbol{Y}$, and $\mathrm{d}\boldsymbol{Z}$. The volume differential in the reference configuration is given by the product $\mathrm{d}V = \left(\mathrm{d}\boldsymbol{X} \times \mathrm{d}\boldsymbol{Y}\right) \cdot \mathrm{d}\boldsymbol{Z}$. Take three infinitesimal vectors in the spatial configuration such that $\mathrm{d}\boldsymbol{x} = \boldsymbol{F}\mathrm{d}\boldsymbol{X}$, $\mathrm{d}\boldsymbol{y} = \boldsymbol{F}\mathrm{d}\boldsymbol{Y}$, and $\mathrm{d}\boldsymbol{z} = \boldsymbol{F}\mathrm{d}\boldsymbol{Z}$. The volume differential in the deformed configuration is the result of $\mathrm{d}v = \left(\mathrm{d}\boldsymbol{x} \times \mathrm{d}\boldsymbol{y}\right) \cdot \mathrm{d}\boldsymbol{z}$. Use Eq. (39) to show the push-forward of the volume differential

$$\mathrm{d}v = J\mathrm{d}V, \quad J = \det \boldsymbol{F}. \tag{103}$$

3.4. Time derivative of functions defined in the spatial configuration

Suppose there is a function defined as $\boldsymbol{f} = \boldsymbol{f}\left(\boldsymbol{x}, t\right)$, $\boldsymbol{f} : \mathbb{R}^3 \times \mathbb{R} \mapsto \mathcal{T}^n$, i.e. it is a mapping of the spatial position vector and time to a n-th order tensor space. Taking the time derivative of such function is not trivial, since the spatial position vector is itself a function of time. Thus, the *total* derivative must be evaluated under the chain rule,

$$\frac{\mathrm{d}\boldsymbol{f}}{\mathrm{d}t} = \dot{\boldsymbol{f}} = \frac{\partial \boldsymbol{f}}{\partial t} + \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \cdot \frac{\partial \boldsymbol{x}}{\partial t} = \frac{\partial \boldsymbol{f}}{\partial t} + \mathrm{grad}\left(\boldsymbol{f}\right) \cdot \boldsymbol{v}. \tag{104}$$

The second term, $\mathrm{grad}\left(\boldsymbol{f}\right) \cdot \boldsymbol{v}$ is known as the *convective term* of the total derivative, for it is a correction to the rate of change of the particle's position in space during deformation.

Challenge 3.2

Using the chain rule, demonstrate that the time derivative of the jacobian $J = \det \boldsymbol{F}$ is given by

$$\frac{\mathrm{d}J}{\mathrm{d}t} = \dot{J} = J\boldsymbol{F}^{-\mathrm{T}} : \dot{\boldsymbol{F}}. \tag{105}$$

Then, using Eq. (95), Eq. (97), and Eq. (102), show that

$$\dot{J} = J\mathrm{div}\left(\boldsymbol{v}\right), \tag{106}$$

such that the time derivative of the jacobian is the jacobian itself multiplied by the spatial divergence of the velocity field.

For a physical intuition of Eq.(104), imagine the particle contained in the wagon of a train. The partial derivative measures the rate of change of the function $\boldsymbol{f}$ with respect to the frame of reference of a point fixed out in the track. The convective term, on the other hand, is the correction due to the movement of the particle relative to the gradient of the field in the region crossed by the wagon at that time. Thus, the convective term reaches maximum when the velocity vector is in the same direction as the last index of the spatial gradient. The opposite direction to the last index of the gradient gives the velocity to which the convective term is minimum. Finally, it is null if the velocity vector is orthogonal to the direction of the last index of the spatial gradient of $\boldsymbol{f}$.

3.5. Integral theorems

This subsection, as was the Subsec. 3.3, is essentially a calculus-oriented piece of text; despite the physics-oriented analogies and explanations. We chose to separate the topic of gradient and divergence operators and the correlated integral theorems from the calculus section. This choice is justified by the concept of material and spatial configurations: there is no sense in discussing configurations in parallel with a pure calculus-section.

We will provide the next expressions in the spatial configuration only, for the sake of keeping this text as objective and straightforward as possible. Furthermore, the reader can prove the extension of the following expressions to the material configuration quite easily.

The *divergence theorem* (also known as the Gauss' divergence theorem) stands the equivalence of the integration in the boundary with the integration in the volume. For a first-order tensor field $\boldsymbol{a}$,

$$\int\limits_{\Gamma} \boldsymbol{a} \cdot \boldsymbol{n} \, \mathrm{d}s = \int\limits_{\Omega} \mathrm{div}\,(\boldsymbol{a})\, \mathrm{d}v, \tag{107}$$

where $\boldsymbol{n}$ is the normal vector to the boundary Γ of the spatial configuration. Conversely for a second-order tensor $\boldsymbol{A}$,

$$\int\limits_{\Gamma} \boldsymbol{A}\boldsymbol{n} \, \mathrm{d}s = \int\limits_{\Omega} \mathrm{div}\,(\boldsymbol{A})\, \mathrm{d}v. \tag{108}$$

Exercise 3.3

Prove Eq. (108) using Einstein notation to equate the multiplication of the second-order tensor $\boldsymbol{A}$ with the spatial normal vector with the contraction of the vectors on the boundary of Eq. (107).

There are other useful iterations of the divergence theorem that can be proved from

Eq. (107), which are

$$
\begin{aligned}
\text{i)} \int\limits_{\Gamma} \phi \boldsymbol{a} \cdot \boldsymbol{n} \, \mathrm{d}s &= \int\limits_{\Omega} \operatorname{div}(\phi \boldsymbol{a}) \, \mathrm{d}v, \quad \phi : \mathbb{R}^3 \mapsto \mathbb{R}, \quad \boldsymbol{a} \in \mathbb{R}^3 \\
\text{ii)} \int\limits_{\Gamma} \phi \boldsymbol{A} \boldsymbol{n} \, \mathrm{d}s &= \int\limits_{\Omega} \operatorname{div}(\phi \boldsymbol{A}) \, \mathrm{d}v, \quad \phi : \mathbb{R}^3 \mapsto \mathbb{R}, \quad \boldsymbol{A} \in \mathcal{T}^2, \\
\text{iii)} \int\limits_{\Gamma} \boldsymbol{a} \cdot \boldsymbol{A} \boldsymbol{n} \, \mathrm{d}s &= \int\limits_{\Omega} \operatorname{div}\left(\boldsymbol{A}^{\mathrm{T}} \boldsymbol{a}\right) \mathrm{d}v, \quad \boldsymbol{a} \in \mathbb{R}^3, \quad \boldsymbol{A} \in \mathcal{T}^2, \\
\text{iv)} \int\limits_{\Gamma} \boldsymbol{a} \otimes \boldsymbol{n} \, \mathrm{d}s &= \int\limits_{\Omega} \operatorname{grad}(\boldsymbol{a}) \, \mathrm{d}v, \quad \boldsymbol{a} \in \mathbb{R}^3.
\end{aligned} \tag{109}
$$

The divergence theorem is particularly useful for analyzing fluxes. We will discuss more about fluxes in Subsec. 5.1 as a prelude to conservation laws. However, to provide the reader with a hint about what is to come, the divergence theorem plays a major role in determining how much of a quantity crosses a boundary evaluating the integration of the divergence in the volume enclosed by such boundary, and vice-versa. Consequently, the divergence theorem is fundamental to deriving partial differential equations from fundamental principles to model physical phenomena, such as heat transfer, fluid dynamics, and elasticity.

Exercise 3.4

1. Prove items i), ii), and iii) of Eq. (109) using Eq. (107) and Eq. (108).
2. Prove item iv) of Eq. (109) using the identity $\operatorname{div}(\cdot) = \operatorname{tr}\big(\operatorname{grad}(\cdot)\big)$ and Eq. (107).

The product rule for derivatives can be applied to the divergence theorem using components. This trick will be quite useful to evaluate time derivatives of spatial quantities and weak formulations for the finite element method. Thus, we can easily show that item i) in Eq. (109) can be reduced to

$$
\int\limits_{\Omega} \operatorname{div}(\phi \boldsymbol{a}) \, \mathrm{d}v = \int\limits_{\Omega} \phi \operatorname{div}(\boldsymbol{a}) + \operatorname{grad}(\phi) \cdot \boldsymbol{v} \, \mathrm{d}v. \tag{110}
$$

Exercise 3.5

Show that Eq. (110) is true. Then, show that the following relation is also true,

$$
\int\limits_{\Omega} \operatorname{div}(\boldsymbol{A}) \cdot \boldsymbol{u} \, \mathrm{d}v = \int\limits_{\Omega} \operatorname{div}\left(\boldsymbol{A}^{\mathrm{T}} \boldsymbol{u}\right) \mathrm{d}v - \int\limits_{\Omega} \boldsymbol{A} : \operatorname{grad}(\boldsymbol{u}) \, \mathrm{d}v, \tag{111}
$$

where $\boldsymbol{A}$ is a second-order tensor and $\boldsymbol{u}$ is a first-order tensor.

Challenge 3.3

Recollect that $\boldsymbol{u} \times \boldsymbol{v} = \boldsymbol{\varepsilon} : (\boldsymbol{u} \otimes \boldsymbol{v})$ and use item ii) of Eq. (109) with $\phi = 1$ to prove that

$$\int_{\Gamma} \boldsymbol{r} \times \boldsymbol{A}\boldsymbol{n} \, \mathrm{d}s = \int_{\Omega} \boldsymbol{\varepsilon} : \mathrm{grad}\,(\boldsymbol{r})\, \boldsymbol{A}^{\mathrm{T}} + \boldsymbol{r} \times \mathrm{div}\,(\boldsymbol{A})\, \mathrm{d}v. \tag{112}$$

3.6. Polar decomposition

Even though the deformation gradient can be used to transform an infinitesimal vector (also known as a *fibril*) in the reference to the deformed configurations (see Eq. (88)),

$$\mathrm{d}\boldsymbol{x} = \boldsymbol{F}\mathrm{d}\boldsymbol{X}, \tag{113}$$

the deformation gradient, $\boldsymbol{F}$, is not a valid strain measure. The answer to this claim is: strain is a measure that must be **unaffected by rotations**. This is straightforward to realize: if a portion of a body simply rotates in space, each particle within that portion does not distance itself with reference to the other particles.

Fortunately, there is a tensor decomposition that rewrites a second-order tensor as the product of a proper orthogonal tensor (a rotation tensor) and a symmetric tensor. This procedure is known as the *polar decomposition*, which states that

$$\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U}, \quad \boldsymbol{F} = \boldsymbol{v}\boldsymbol{R}; \quad \boldsymbol{R} \in \mathrm{SO}^3, \ \boldsymbol{U}, \boldsymbol{v} \in \mathrm{Sym}^3. \tag{114}$$

The polar decomposition is unique if $\det \boldsymbol{F} > 0$ [1, 7]. The polar decomposition of the deformation gradient tensor gives physical meaning to the symmetric tensors of the decomposition—$\boldsymbol{U}$ is the *material stretch tensor*, i.e. it signals the principal directions and the principal values of stretch in the material (reference) configuration; whereas $\boldsymbol{v}$ signals the principal directions and the principal values of stretch in the spatial (deformed) configuration.

Suppose that you are handling a piece of dough, you might be kneading a loaf of bread. When you *stretch* that piece of dough holding one hand down to the table and pushing the other hand away from yourself, particles in the dough aligned with the **direction** linking your far-positioned hand to you will become further away from each other, the distances between them will increase. In contrast, particles aligned to a **direction** perpendicular to the first one will become closer to each other, for the width of the dough piece will decrease, thus, the distances between particles in that direction will also decrease.

The mental experiment just shown illustrates the concept of principal directions, i.e the directions where the particles move the most relative to each other; and the concept of principal stretches, which translates to the ratio of the distances between particles from the current state to the initial state. Hence, principal stretches cannot be negative, since distances are strictly positive and so will be ratios of positive quantities. Additionally, a ratio greater than 1 means tension, as the particles being pulled further apart in the direction away from you in the dough-kneading example; and a ratio between 0 and 1 means compression, the particles are being pushed together, as in the direction perpendicular to the kneading direction, in the example.

We can rewrite Eq. (114) to equate the stretch tensors,

$$\boldsymbol{R}\boldsymbol{U} = \boldsymbol{v}\boldsymbol{R} \implies \boldsymbol{v} = \boldsymbol{R}\boldsymbol{U}\boldsymbol{R}^{\mathrm{T}} \text{ and } \boldsymbol{U} = \boldsymbol{R}^{\mathrm{T}}\boldsymbol{v}\boldsymbol{R}. \tag{115}$$

From Subsubsec. 1.5.10 and due to the fact that the rotation tensor is an orthogonal tensor, it is clear that the stretch tensors are related to each other by a similarity transformation and thus possess the same eigenvalues. As a matter of fact, the eigenvectors of the spatial stretch tensor, denoted as $\boldsymbol{n}_i \ \ \forall i \in \{1,2,3\}$, are merely the result of rotation of the eigenvectors of the material stretch tensor, $\boldsymbol{N}_i \ \ \forall i \in \{1,2,3\}$. Consequently, Equation (115) presents the push-forward operation from the material stretch tensor to the spatial stretch tensor, and the pull-back in the reverse direction.

Physically, the eigenvectors of the stretch tensors have the same meaning as the principal directions we discussed in the dough-kneading mental experiment. Conversely, the eigenvalues of the stretch tensors have the same meaning of that experiment—ratio of distances in the current to the initial states between a particle to its infinitesimally immediate neighbor particle along the corresponding principal direction.

Exercise 3.6

Using the spectral decomposition of the stretch tensors and Eq. (115), show that

i) the spatial principal directions of stretch, $\boldsymbol{n}_i$, are a rotation of the material principal directions of stretch, $\boldsymbol{N}_i$;

ii) the eigenvalues of the stretch tensors are the same and, thus, stretch in the principal directions have the same value in both configurations.

If we plug Eq. (114) into the equation for the relation between two infinitesimal vectors in both configurations, Eq. (113), we will have two options,

$$\begin{aligned} &\text{i) } \mathrm{d}\boldsymbol{x} = \boldsymbol{R}\boldsymbol{U}\mathrm{d}\boldsymbol{X}, \\ &\text{ii) } \mathrm{d}\boldsymbol{x} = \boldsymbol{v}\boldsymbol{R}\mathrm{d}\boldsymbol{X}. \end{aligned} \tag{116}$$

Equation (116) shows a subtle difference in using the two facets of the polar decomposition. This difference carries a profound physical interpretation and, to fully comprehend it, we have to step back to the interpretation of the deformation gradient tensor itself. Equation (116) i) uses the *right polar decomposition*, since the symmetric tensor is to the right; whereas Equation (116) ii) deploys the *left polar decomposition*, since the symmetric tensor is to the left of the decomposition.

In Eq. (113), the deformation gradient tensor transforms a vector differential in the reference configuration to a vector differential in the deformed configuration. Thus, the deformation gradient contains all possible deformation modes—it both stretches and rotates the vector infinitesimal; however, with no particular order. Armed with the polar decomposition, in contrast, we can indeed enforce an order of the transformation kinematics. Equation (116) i), for instance, stretches the vector differential, then rotates it. Equation (116) ii), on the other hand, rotates the vector differential first and then stretches it.

3.6.1. Material, spatial, and two-point tensors

When we say a tensor is *material*, such as the material stretch tensor and the material position vector, we mean that *all components* of the tensor are related to quantities measured or derived in the reference configuration. When a tensor is said to be *spatial*, such as the spatial stretch tensor and the spatial position vector, the opposite happens—all components of the tensor are related to quantities measured or derived in the spatial configuration.

Concerns about the nature of a tensor (material or spatial) are paramount, since contraction operations cannot be performed between vectors of different vector spaces (configurations). Otherwise, by absurd and sake of argument, if contraction could be carried out between vectors of different vector spaces, such operation would not be an inner product (see Subsec. 1.6).

Nonetheless, there is another kind of tensor: the *two-point tensor*. Each component of a two-point tensor is related to a quantity that is measured or derived using information from both quantities. One such example is the deformation gradient: the spatial position vector is differentiated with respect to the corresponding material position vector, thus, information of both configurations are present and used.

We might also discern the nature of a tensor by and to its indices. To illustrate this claim, take the deformation gradient: due to our convention of right-differentiation, the information coming from the material configuration is stacked to the right and, thus, *second index is material.* Conversely, the information coming from the spatial configuration stands to the left, at the first index. In summary, the deformation gradient is an example of two-point tensor whose first index is spatial and the second index is material.

Given the constraint on contraction operators to contract indices of the same nature only, it is once again clear why the deformation gradient has a material second index. As the deformation gradient tensor transforms infinitesimal vectors in the reference configuration to infinitesimal vectors in the spatial configuration, the second index must be material to contract with the material vector; whereas the first index must be spatial to return a spatial vector as result.

Due to the polar decomposition, it is clear that the rotation tensor $\boldsymbol{R}$ in Eq. (114) is also a two-point tensor, since both stretch tensors are purely material or spatial. By inspection of Eq. (114) and Eq. (115), one shall infer that the rotation tensor, much like the deformation gradient tensor, has the first index as spatial and the second index as a material index.

We will take care about the nature of the indices of all tensors in the next sections and, thus, the reader is invited to come back to this subsubsection whenever you see fit.

3.7. Strain tensors

The most intuitive visualization of strain is the uniaxial case, where a sufficiently long and thin body, such as a string, is stretched. As the deformation of each particle seems to be along the loading axis, the strain seems to be equivalent to the stretch, i.e. a ratio of of the initial to the final distances between particles. We might, however, extend this visualization to the three-dimensional case by evaluating the inner product of an infinitesimal vector in the deformed configuration.

The inner product, as discussed in Subsec. 1.6, can be used to *generate a norm*. Thus, it can be used to evaluate lengths and distances, just as the measurement of stretch requires. Moreover, we can correlate this measurement to material quantities,

$$\mathrm{d}\boldsymbol{x} \cdot \mathrm{d}\boldsymbol{x} = (\boldsymbol{F}\mathrm{d}\boldsymbol{X}) \cdot (\boldsymbol{F}\mathrm{d}\boldsymbol{X}) = \mathrm{d}\boldsymbol{X} \cdot \left(\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}\mathrm{d}\boldsymbol{X}\right) = \mathrm{d}\boldsymbol{X} \cdot (\boldsymbol{C}\mathrm{d}\boldsymbol{X}), \tag{117}$$

where $\boldsymbol{C} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}$ is known as the *right Cauchy-Green* deformation tensor. The right Cauchy-Green deformation tensor is symmetric positive-definite. To prove that we shall use the right polar decomposition,

$$\boldsymbol{C} = \boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} = (\boldsymbol{R}\boldsymbol{U})^{\mathrm{T}}\boldsymbol{R}\boldsymbol{U} = \boldsymbol{U}^{\mathrm{T}}\boldsymbol{R}^{\mathrm{T}}\boldsymbol{R}\boldsymbol{U} = \boldsymbol{U}^{\mathrm{T}}\boldsymbol{U} = \boldsymbol{U}^2 \implies \boldsymbol{U} = \sqrt{\boldsymbol{C}}. \tag{118}$$

As the right Cauchy-Green deformation tensor is a power of the material stretch tensor, $\boldsymbol{C}$ and $\boldsymbol{U}$ share the same eigenspace. Note that the rotation component of the deformation gradient tensor has completely vanished from the right Cauchy-Green deformation tensor, thus, the latter is *rotation-invariant*.

Remark 3.4

As $\boldsymbol{C}$ defines an inner product to define lengths and angles in the spatial configuration from vectors in the reference configuration, the right Cauchy-Green is the *metric tensor* corresponding to the transformation from the material to the spatial configurations. Yet another connection between continuum mechanics and differential geometry.

Recall from Eq. (91) that, in the case of a rigid body kinematics, the deformation gradient can be the identity, and so will be the right Cauchy-Green tensor. Thus, $\boldsymbol{C}$ is not null where there is no strain involved and that is the reason why it is called a *deformation* tensor; not a *strain* tensor. To circumvent this, we alter the inner product equation by subtracting the length in the reference configuration,

$$\frac{\mathrm{d}\boldsymbol{x}\cdot\mathrm{d}\boldsymbol{x}-\mathrm{d}\boldsymbol{X}\cdot\mathrm{d}\boldsymbol{X}}{2}=\frac{\mathrm{d}\boldsymbol{X}\cdot(\boldsymbol{C}\mathrm{d}\boldsymbol{X})-\mathrm{d}\boldsymbol{X}\cdot(\boldsymbol{I}\mathrm{d}\boldsymbol{X})}{2}=\mathrm{d}\boldsymbol{X}\cdot\left(\frac{1}{2}\left[\boldsymbol{C}-\boldsymbol{I}\right]\mathrm{d}\boldsymbol{X}\right)$$
$$=\mathrm{d}\boldsymbol{X}\cdot(\boldsymbol{E}\mathrm{d}\boldsymbol{X})\implies\boldsymbol{E}=\frac{1}{2}\left[\boldsymbol{C}-\boldsymbol{I}\right]; \tag{119}$$

where $\boldsymbol{E}$ is known as the *Green-Lagrange strain tensor*. Due to the shift by the identity tensor, the Green-Lagrange strain tensor is indeed null when there is only rigid-body motion, both of the whole body or of the particle with respect to the immediate infinitesimal vicinity.

Equation (119) can be rewritten transforming the material infinitesimal vector to the spatial configuration, using the inverse of relation in Eq. (113), i.e. $\mathrm{d}\boldsymbol{X}=\boldsymbol{F}^{-1}\mathrm{d}\boldsymbol{x}$,

$$\frac{\mathrm{d}\boldsymbol{x}\cdot\mathrm{d}\boldsymbol{x}-\mathrm{d}\boldsymbol{X}\cdot\mathrm{d}\boldsymbol{X}}{2}=\frac{\mathrm{d}\boldsymbol{x}\cdot\mathrm{d}\boldsymbol{x}-\left(\boldsymbol{F}^{-1}\mathrm{d}\boldsymbol{x}\right)\cdot\left(\boldsymbol{F}^{-1}\mathrm{d}\boldsymbol{x}\right)}{2}=\mathrm{d}\boldsymbol{x}\cdot\left(\frac{1}{2}\left[\boldsymbol{I}-\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{-1}\right]\mathrm{d}\boldsymbol{x}\right)$$
$$=\mathrm{d}\boldsymbol{x}\cdot\left(\frac{1}{2}\left[\boldsymbol{I}-\left(\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}\right)^{-1}\right]\mathrm{d}\boldsymbol{x}\right)=\mathrm{d}\boldsymbol{x}\cdot\left(\frac{1}{2}\left[\boldsymbol{I}-\boldsymbol{b}^{-1}\right]\mathrm{d}\boldsymbol{x}\right)=\mathrm{d}\boldsymbol{x}\cdot(\boldsymbol{e}\mathrm{d}\boldsymbol{x});$$
$$\boldsymbol{b}=\boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}=\boldsymbol{v}^2,\quad \boldsymbol{e}=\frac{1}{2}\left[\boldsymbol{I}-\boldsymbol{b}^{-1}\right]. \tag{120}$$

The tensor $\boldsymbol{b}$ is known as the *left Cauchy-Green deformation tensor*. Whereas $\boldsymbol{e}$ is known as the *Euler-Almansi strain tensor*. The $\boldsymbol{b}$ is the transposition of the right Cauchy-Green tensor and, thus, suffers from the same ailment as the latter-it is the identity tensor for rigid-body motion. Consequently, the Euler-Almansi strain tensor uses the same trick to become null for rigid-body kinematics.

The left Cauchy-Green and the Euler-Almansi tensors are purely spatial tensors, in contrast to $\boldsymbol{C}$ and $\boldsymbol{E}$, which are purely material tensors. Thereby, we can derive the push-forward operation from the Green-Lagrange to the Euler-Almansi strain tensors,

$$\boldsymbol{e}=\frac{1}{2}\left[\boldsymbol{I}-\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{-1}\right]=\frac{1}{2}\left[\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}\boldsymbol{F}^{-1}-\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{F}^{-1}\right]=\frac{1}{2}\boldsymbol{F}^{-\mathrm{T}}\left[\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F}-\boldsymbol{I}\right]\boldsymbol{F}^{-1}$$
$$=\frac{1}{2}\boldsymbol{F}^{-\mathrm{T}}\left[\boldsymbol{C}-\boldsymbol{I}\right]\boldsymbol{F}^{-1}=\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{E}\boldsymbol{F}^{-1}\implies\boldsymbol{e}=\boldsymbol{F}^{-\mathrm{T}}\boldsymbol{E}\boldsymbol{F}^{-1}. \tag{121}$$

Conversely, the pull-back from the spatial to the material configuration is given by

$$\boldsymbol{E}=\boldsymbol{F}^{\mathrm{T}}\boldsymbol{e}\boldsymbol{F}. \tag{122}$$

Exercise 3.7

Using the spectral decomposition of the stretch tensors, show that

i) the eigenvalues of the right Cauchy-Green and of the left Cauchy Green deformation tensors are equal to the square root of the eigenvalues of their respective stretch tensors;

ii) the right Cauchy-Green and the left Cauchy-Green deformation tensors share the same invariants.

Exercise 3.8

1. Use identity $\boldsymbol{F}^{-1}\boldsymbol{F} = \boldsymbol{I}$ to show that the inverse of a tensor *invert the nature of its indices.* In this case, the second index becomes spatial and the first one material.

2. Use the polar decomposition to demonstrate that the transposition $\boldsymbol{F}^{\mathrm{T}}$ inverts the nature of the tensor indices just as the inverse operation does.

3. Use Einstein notation to show that the push-forward and pull-back operations defined in Eq. (121) and in Eq. (122) involve contractions of the indices of the same nature only (spatial with spatial and material with material).

4. The *infinitesimal strain tensor* is given by $\boldsymbol{\varepsilon} = \mathrm{sym}\big(\mathrm{Grad}\,(\boldsymbol{u})\big)$. Show that this strain tensor is indeed affected by rotations.

4. Tractions and stress tensors

In Sec. 3, we discussed how deformable bodies move; or, even more precisely, we defined a mathematical framework to model how each particle in this body moves in space. Due to Newton's laws of motion, we know that change in movement (acceleration) is the product of forces; in the other hand, the lack of movement might also be the result of forces that balance each other out. In summary, it is clear that we must include forces in the framework we posed.

Forces usually appeared in continuum mechanics as a distribution on a surface, i.e. there is not a single force vector on a point in the body; rather a function that spreads the force over a surface. Thus, we reach out to a new concept—*traction*, the ratio of force per unit area in the limit of an infinitesimal area element,

$$\boldsymbol{t} = \lim_{\mathrm{d}A \to 0} \frac{\mathrm{d}\boldsymbol{f}}{\mathrm{d}A} \quad \left[\mathrm{N/m^2}\right] = [\mathrm{Pa}]\,. \tag{123}$$

It is important to note that tractions emerge on *boundaries.* Hence, they are forces distributed over a body external boundary or inside it if we create a new boundary by slicing the body into two or more pieces. There are two readily available examples of tractions in nature: pressure and friction. For example, a fluid exerts pressure (a distributed force normal to the surface) on the surface of its container; and a rod exerts friction (a distributed force tangent to the surface) on its bearing.

To illustrate pressure and friction tractions, place your indicating finger onto the back of your thumb. Then, first press the indicating finger against your thumb; you will feel a force normal to your thumb spread across the contact surface of the two fingers, this is pressure. Now, slide your indicating finger on your thumb away from your knuckle; you will feel the skin of your indicating finger pulling the skin of your thumb tangentially, which is an example of friction. This way, it is clear that traction is a vector with direction.

Returning to a generic body, such as that of Fig. 1. This body might be subject to tractions on its boundary, pulling and pushing it. Additionally, there might be *body forces* distributed in the domain (volume) of the body, i.e. forces given by force volumetric densities ($\left[\mathrm{N/m^3}\right]$). Body forces are fields in the body domain. The Euler-Cauchy stress principle states that if we slice the body in two pieces, the action of the forces in the section of the original boundary and the action of the body forces in the section of the original volume owned by one of the pieces equals the action of the forces acting in the newly created boundary. Moreover, the traction acting on the new boundary of one piece will be reciprocal to the traction acting on the adjacent piece, hence the tractions on adjacent bodies have the same norm but opposite directions.

Note that the traction on a newly created boundary by slicing depends upon the orientation of the slicing. Imagine a long rod pulled by its ends, if you slice it perpendicular to its axis, the traction in the newly crated boundary will be mostly aligned with the rod longitudinal axis. In contrast, if you slice this rod in a tilted direction, components of traction tangent to the newly generated boundary will appear. Thus, the traction vector depends on the slicing orientation as well as its position relative to the body. Nonetheless, the *Cauchy postulate* states the traction on a point depends upon a single information of the boundary surface—its point-wise normal vector, $\boldsymbol{n}$. Consequently, the traction vector does not depend on the local curvature of the surface evaluated at the point.

4.1. Cauchy stress tensor

To this point, we realized that the traction vector is a function of the position of a point on a surface in the body and of the point-wise normal vector of the said surface. We also observed that the sum of the action of all forces, including those generated by tractions, must balance each other or be equal to the acceleration of the body. What is missing is how to relate the traction vector with the surface normal vector. This is precisely the objective of the *Cauchy stress theorem*.

Suppose we take an infinitesimal chunk of material away of a body. This chunk has the particular shape of a tetrahedron, with three mutually orthogonal faces that are aligned with the basis vectors. The fourth facet is slanted with reference to the other facets and has a

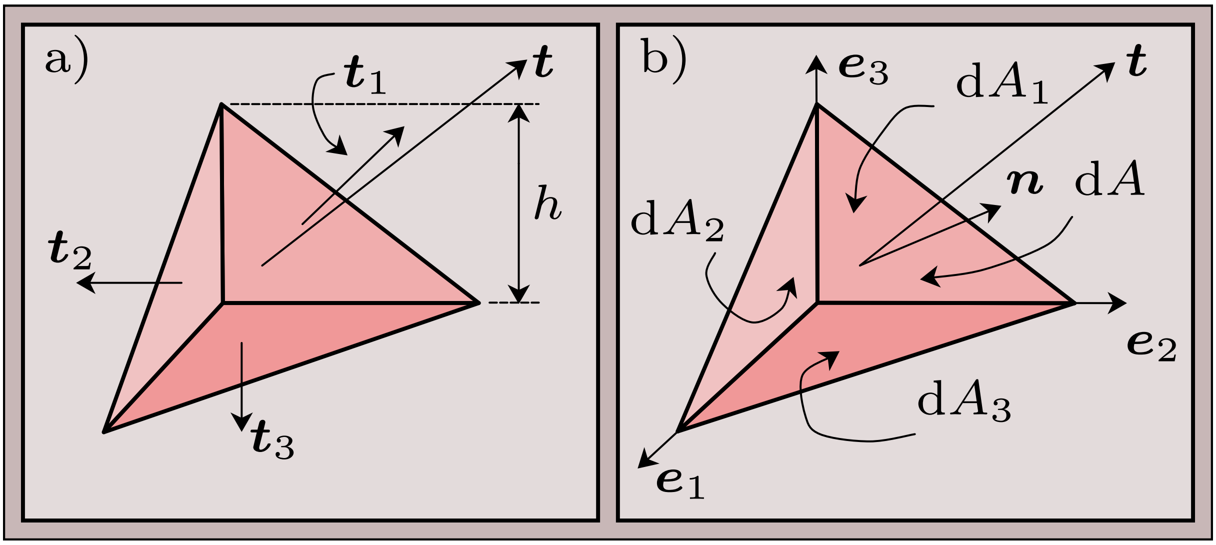

Figure 2: The Cauchy tetrahedron. A tetrahedron with three mutually orthogonal facets that are aligned with the basis vectors. The volume of the Cauchy tetrahedron and the area of each facet are infinitesimal. The facet with normal vector $\boldsymbol{n}$ is the *slanted facet*, where the unknown traction $\boldsymbol{t}$ is applied. Frame a) shows the tractions on the facets; and frame b) illustrates the different area differentials and the overall orientation of the tetrahedron.

generic normal vector $\boldsymbol{n}$. Each facet has an infinitesimal surface area nominated accordingly to the facet number, see Fig. 2. The balance of forces (second Newton's law of motion) in the tetrahedron is given by

$$-\boldsymbol{t}_1\mathrm{d}A_1 + -\boldsymbol{t}_2\mathrm{d}A_2 + -\boldsymbol{t}_3\mathrm{d}A_3 + \boldsymbol{t}\mathrm{d}A = \mathrm{d}m\ddot{\boldsymbol{u}} = \frac{\rho}{3}h\mathrm{d}A\left(\boldsymbol{n}\cdot\boldsymbol{e}_3\right)\ddot{\boldsymbol{u}}, \tag{124}$$

where $\mathrm{d}m$ is the differential of mass of the matter occupying the tetrahedron; ρ is the density of the said matter; $\ddot{\boldsymbol{u}}$ is the second time-derivative of the displacement vector of the tetrahedron in space.

One should note that the numbered area differentials are projections of the area differential corresponding to the slanted facet, such that

$$\mathrm{d}A_1 = \mathrm{d}A\left(\boldsymbol{n}\cdot\boldsymbol{e}_1\right), \quad \mathrm{d}A_2 = \mathrm{d}A\left(\boldsymbol{n}\cdot\boldsymbol{e}_2\right), \quad \mathrm{d}A_3 = \mathrm{d}A\left(\boldsymbol{n}\cdot\boldsymbol{e}_3\right). \tag{125}$$

Hence, we might substitute Eq. (125) into Eq. (124) and, then, divide both sides of the latter by $\mathrm{d}A$,

$$-\boldsymbol{t}_1\left(\boldsymbol{n}\cdot\boldsymbol{e}_1\right) - \boldsymbol{t}_2\left(\boldsymbol{n}\cdot\boldsymbol{e}_2\right) - \boldsymbol{t}_3\left(\boldsymbol{n}\cdot\boldsymbol{e}_3\right) + \boldsymbol{t} = \frac{\rho}{3}h\mathrm{d}A\left(\boldsymbol{n}\cdot\boldsymbol{e}_3\right)\ddot{\boldsymbol{u}}. \tag{126}$$

If the limit of the characteristic size of the tetrahedron, h, is taken with $h \to 0$, Equation (126) loses the influence of the acceleration force and, thus, can be rewritten to

$$\boldsymbol{t} = \boldsymbol{t}_1 \otimes \boldsymbol{e}_1 \cdot \boldsymbol{n} + \boldsymbol{t}_2 \otimes \boldsymbol{e}_2 \cdot \boldsymbol{n} + \boldsymbol{t}_3 \otimes \boldsymbol{e}_3 \cdot \boldsymbol{n} = \Big[\big(t_i^1 \boldsymbol{e}_i \otimes \boldsymbol{e}_1\big) + \big(t_i^2 \boldsymbol{e}_i \otimes \boldsymbol{e}_2\big) + \big(t_i^3 \boldsymbol{e}_i \otimes \boldsymbol{e}_3\big)\Big] \cdot \boldsymbol{n}$$
$$= \sigma_{ij} \boldsymbol{e}_i \otimes \boldsymbol{e}_j \cdot n_k \boldsymbol{e}_k \implies \boldsymbol{t} = \boldsymbol{\sigma} \boldsymbol{n}. \tag{127}$$

The tensor $\boldsymbol{\sigma}$ is known as the *Cauchy stress tensor*, and it relates normal vectors defining slices or boundaries with the corresponding traction vector. We showed that this tensor exists and that it is uniquely given by the traction vector. Furthermore, we could demonstrate that the relation between the normal vector and the traction vector is a linear mapping. Keep in mind that the Cauchy stress tensor is a spatial tensor, for it relates normal vectors in the spatial configuration with traction vectors in the same configuration.

4.2. First Piola-Kirchhoff stress tensor

In Subsec. 3.7, we discussed strain tensors and, there, naturally appeared material and spatial tensors. Additionally, we provided operations between configurations, the now well-acquainted push-forward and pull-back operations. The immediate question is if we could do the same for stress. The answer is positive, but first we need to define what is a traction vector in the reference configuration.

As we discussed before, the concept of the reference configuration arises to facilitate integration, since the integration domain remains unchanged in the material configuration. Looking towards integration we define a *material traction vector* $\boldsymbol{T}$ such that

$$\boldsymbol{T}\colon\ \boldsymbol{T}\mathrm{d}S = \boldsymbol{t}\mathrm{d}s, \tag{128}$$

where $\mathrm{d}S$ is the area differential in the reference configuration and $\mathrm{d}s$ is the are differential in the deformed configuration. One might realize that this postulate of a traction vector strongly enforces that the force differential is the same in both configurations. As the area differentials are scalars, the material and spatial traction vectors are collinear, but they possess different magnitudes.

We postulate as well there is a second-order tensor that is an analogue to the Cauchy stress theorem, but in the material configuration,

$$\boldsymbol{T} = \boldsymbol{P}\boldsymbol{N} \implies \boldsymbol{T}\mathrm{d}S = \boldsymbol{P}\boldsymbol{N}\mathrm{d}S = \boldsymbol{P}\mathrm{d}\boldsymbol{S}, \tag{129}$$

in which $\boldsymbol{N}$ is the normal vector to a surface evaluated on a point in the reference configuration; $\boldsymbol{P}$ is the *first Piola-Kirchhoff stress tensor*; and $\mathrm{d}\boldsymbol{S}$ is the outwardly-oriented area differential. The postulates in Eq. (128) and Eq. (129), unfortunately, do not suffice, because we need to correlate the area differentials to recover a closed-form expression. This is the feature and purpose of the *Nanson's formula*, a rule for transforming surface differentials between configurations.

4.2.1. Nanson's formula

We recollect that the volume differential in both configuration can be related to the oriented area differentials by the inner product with an infinitesimal vector (the oriented area differential and the infinitesimal vector form a three-dimensional hexahedron),

$$\mathrm{d}V = \mathrm{d}\boldsymbol{Z} \cdot \mathrm{d}\boldsymbol{S}, \;\; \mathrm{d}v = \mathrm{d}\boldsymbol{z} \cdot \mathrm{d}\boldsymbol{s}. \tag{130}$$

The push-forward of the volume differential from the material to the spatial configurations (see Eq.(103)) can be applied to Eq. (130) alongside the relation $\mathrm{d}\boldsymbol{z} = \boldsymbol{F}\mathrm{d}\boldsymbol{Z}$ to yield

$$\begin{aligned} \mathrm{d}v = J\mathrm{d}V \implies (\boldsymbol{F}\mathrm{d}\boldsymbol{Z}) \cdot \mathrm{d}\boldsymbol{s} = \mathrm{d}\boldsymbol{Z} \cdot \left(\boldsymbol{F}^{\mathrm{T}}\mathrm{d}\boldsymbol{s}\right) &= J\mathrm{d}\boldsymbol{Z} \cdot \mathrm{d}\boldsymbol{S} \\ \implies \mathrm{d}\boldsymbol{Z} \cdot \left[\boldsymbol{F}^{\mathrm{T}}\mathrm{d}\boldsymbol{s} - J\mathrm{d}\boldsymbol{S}\right] &= 0 \;\; \forall \mathrm{d}\boldsymbol{Z} \in \mathbb{R}^3. \end{aligned} \tag{131}$$

As the projection of the infinitesimal vector onto the term inside brackets must be null for **any** infinitesimal vector, the only possible solution is for the content in the brackets to be the null vector. Henceforth, we can recompile Eq. (131) to the Nanson's formula,

$$\mathrm{d}\boldsymbol{s} = J\boldsymbol{F}^{-\mathrm{T}}\mathrm{d}\boldsymbol{S}, \tag{132}$$

which is the *push-forward of a surface area differential* from the material to the spatial configurations.

The Nanson's formula can be directly applied to Eq. (128), which is combined to Eq. (129),

$$\begin{aligned} \boldsymbol{T}\mathrm{d}S = \boldsymbol{P}\mathrm{d}\boldsymbol{S} = \boldsymbol{t}\mathrm{d}s = \boldsymbol{\sigma}\boldsymbol{n}\mathrm{d}s = \boldsymbol{\sigma}\mathrm{d}\boldsymbol{s} &= J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}\mathrm{d}\boldsymbol{S} \\ \implies \boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}} \text{ and } \boldsymbol{\sigma} &= \frac{1}{J}\boldsymbol{P}\boldsymbol{F}^{\mathrm{T}}. \end{aligned} \tag{133}$$

The first Piola-Kirchhoff stress is a two-point tensor. The nature of $\boldsymbol{P}$ is clear on its transformation from the Cauchy stress tensor—observe that only the second index was transformed, from the spatial to the material configurations. Thus, Equation (129) is a mapping from a normal vector in the reference configuration to a traction vector in the deformed configuration. The physical meaning of the first Piola-Kirchhoff stress is the tensor that provides a traction using a limit close to Eq. (123), with the only difference that the area in the limit is an element of the material configuration.

As the first Piola-Kirchhoff is a two-point tensor, the second Piola-Kirchhoff stress tensor was proposed,

$$\boldsymbol{S} = \boldsymbol{F}^{-1}\boldsymbol{P} = J\boldsymbol{F}^{-1}\boldsymbol{\sigma}\boldsymbol{F}^{-\mathrm{T}}. \tag{134}$$

The inverse of the deformation gradient tensor to the right *pulls-back* the first index to the material configuration. As result, the second Piola-Kirchhoff stress is a material tensor. However, it does not feature much of a physical meaning and does not produce tractions. The second Piola-Kirchhoff stress tensor will be used in computational mechanics within calculations of the finite element method.

5. Conservation laws

Since the dawn of physics, conservation laws have been proposed, axiomatically at first, to model a multitude of phenomena. First, came the conservation of momentum, due to Newton, then the conservation of *action* by Maupertuis and energy by Euler. Later, whole conservation-based formulations were proposed, such as the lagrangian functional by Lagrange and the hamiltonian functional by Hamilton. Noether, a collaborator of illustrious mathematician David Hilbert, published her Noether's theorem in the 20^{th} century, which finally and rigorously stated conservation postulates as symmetries in mathematics.

The scope of this text, nevertheless, is not to introduce excerpts of history of physics or mathematics. We introduced this brief conversational prose to place the reader, and their previous knowledge, in perspective with the works of giants that led and contributed to continuum mechanics.

To derive conservation laws, we will use the spatial configuration, since this configuration is the *true* one, where the physical phenomenon is actually placed. For this reason, we will use the domain Ω and the boundary Γ of the deformed configuration.

5.1. Fluxes and projection onto the boundary

To understand conservation of any physical quantity, one has to understand how this quantity flows in and out of the domain, through its boundary. The boundary is always a topological object with a topological dimension less than the domain; in our case, as the domain lies in three-dimensional space, the boundary is a two-dimensional surface. Again, relying on differential geometry, we locally (point-wise) characterize a surface by its normal vector, i.e. the unitary vector orthogonal to the surface at a point that defines a tangent plane on that point. We also use the convention that the normal vector points outwards and the opposite direction points to the interior of the domain.

Most people associate *flux* with fluids; although this is correct, it is incomplete. Flux means the rate of transport of some physical quantity—matter, energy, heat, etc—moving through a boundary or slice, and hence, it is more precise to say *flux of something*. With this wording, it seems we just moved the problem to another place, to fix it we define *rate of transport* as the time derivative of the quantity being moved or measured.

With that definition of flux, we realize that something is moving along a direction, but this direction might be decomposed in the sum of a portion contained within the plane of the boundary and another normal to it. At this point, one should realize that the direction contained within the plane of the boundary does not leave or enter the domain, rather stays drifting on the boundary. Consequently, the portion of the flux moving in a direction tangent to the boundary plane is worthless for conservation analysis.

Retreating back to Subsubsec. 1.4.1, we recollect that the simple contraction between two vectors is a projection of one onto the other. Hence, we might project the flux direction (vector) onto the normal vector to yield the portion of that direction that is actually leaving or entering the boundary. Finally, the contribution of each point in the boundary can be accounted for using integration over the boundary,

$$\bar{\boldsymbol{a}} = \int_{\Gamma} \boldsymbol{a} \cdot \boldsymbol{n} \, \mathrm{d}s, \tag{135}$$

in which $\bar{\boldsymbol{a}} \in \mathcal{T}^{n-1}$ is the $(n-1)$-order *net flux* tensor and $\boldsymbol{a} \in \mathcal{T}^{n}$ is a generic n-th order flux tensor. We chose to use generic flux and net flux because many physical quantities use the same pattern as Eq. (135).

We adopted the convention that a positive net flux means that the quantity in $\boldsymbol{a}$ is transported out of the domain, thus, a flux vector whose inner product with the spatial normal vector is strictly positive is *leaving* the domain through the boundary, as the normal vector points outwards. This choice of convention is important to set if the net flux must be summed or subtracted in the conservation law.

5.2. Conservation law pattern

Conservation laws translate physical and geometrical intuition into mathematically rigorous equations. These equations, often known as *balance equations*, share a common blueprint: the rate of change of the said quantity in the whole domain must be equal to the amount that enters and leaves through the boundary and the amount that is generated or vanquished in the domain.

The amount that enters and leaves through the boundary is completely accounted by the net flux of the said quantity, as we discussed in Subsec. 5.1. The rate of change of the said quantity in the whole domain is simply denoted by the time derivative of it. Thereby, the only remaining ingredient of a balance equation to discuss in more details is the amount generated and vanquished in the domain.

Generation is point-wise interpreted as a *source*, i.e. there is something coming out of it and becoming part of the domain. Vanquishing, on the other hand, is the point-wise analogue of a *sink*, which translates to something going away from that domain through that point. Sources are conventionalized to have a positive sign, so that they add up to the total of that quantity. Sinks are, in contrast, represented by a negative sign, to subtract from the total. In summary, sources and sinks are two facets of the same coin. As sources and sinks are point-wise objects, we have to integrate over the domain to retrieve the *net generation*,

$$\bar{\boldsymbol{q}} = \int\limits_{\Omega} \boldsymbol{q} \, \mathrm{d}v, \tag{136}$$

where $\boldsymbol{q}$ is the point-wise generation of a physical quantity and, in turn, a volume density of the global quantity $\bar{\boldsymbol{q}}$.

With all ingredients of a balance equation sufficiently well-presented, we can proceed with writing a generic balance equation for a generic physical quantity $\bar{\boldsymbol{c}}$ evaluated at the whole domain,

$$\frac{\mathrm{d}\bar{\boldsymbol{c}}}{\mathrm{d}t} = \bar{\boldsymbol{q}} - \bar{\boldsymbol{a}} = \int\limits_{\Omega} \boldsymbol{q} \, \mathrm{d}v - \int\limits_{\Gamma} \boldsymbol{a} \cdot \boldsymbol{n} \, \mathrm{d}s. \tag{137}$$

Note that we can apply the divergence theorem of Eq. (108) to the term related to net flux in Eq. (137), which enables us to evaluate both integrals in the domain. As a consequence, the generation and the flux portions can be combined into a single integration,

$$\frac{\mathrm{d}\bar{\boldsymbol{c}}}{\mathrm{d}t} = \int\limits_{\Omega} \boldsymbol{q} \, \mathrm{d}v - \int\limits_{\Omega} \operatorname{div}(\boldsymbol{a}) \, \mathrm{d}v = \int\limits_{\Omega} \boldsymbol{q} - \operatorname{div}(\boldsymbol{a}) \, \mathrm{d}v. \tag{138}$$

Observe that we used the total derivative of the global quantity $\bar{\boldsymbol{c}}$ instead of the partial derivative. This choice is motivated by the fact that $\bar{\boldsymbol{c}}$ will be the result of the integral of a function that depends upon the position vector **and** time. Thus, the derivative must be taken as the sum of the partial derivative of $\bar{\boldsymbol{c}}$ with respect to time only plus the chain rule of the partial derivative of the integrated function with respect to the position vector and the dependence of the latter upon time. In contrast, if $\bar{\boldsymbol{c}}$ were a function of time only, the total and partial derivatives would be equal to the other.

Equation (138) presents the basic structure of the majority of conservation laws in classical mechanics in *integral form*. If we consider the name of the form, the predicate *integral* is obvious due to the operation of integration in Eq.(138); but it reveals a much deeper meaning: the integral form accounts for the influence of everything in the whole domain at once, and the integral form does not use any Taylor approximation or hypothesis of a point-wise and sufficiently small chunk of material.

5.3. Reynold's transport theorem

The tensor $\bar{\boldsymbol{c}}$ in Eq. (138) is not a field defined in the domain Ω, but a single value evaluated on the whole domain, just like $\bar{\boldsymbol{q}}$ and $\bar{\boldsymbol{a}}$. Although $\bar{\boldsymbol{c}}$ could indeed be the integration over the domain of a volumetric density $\boldsymbol{c}$ of this quantity. Then, the time-derivative in Eq.

(138) would become

$$\frac{\mathrm{d}\bar{\boldsymbol{c}}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\int\limits_{\Omega} \boldsymbol{c}\,\mathrm{d}v. \tag{139}$$

In conventional calculus courses, it is taught that, when dealing with the derivative of an integral such as Eq. (139), the derivative can be *carried into the integral sign.* In a more rigorous mathematical language, integration and differentiation operators should commute; or, in plain laymen language, it does not matter if the integrand is differentiated first and then integrated, or vice-versa.

To analyze the claim of commutativity between differentiation and integration operators, we should get back to the Riemann integral using the limit of the Riemann sum (see [8]) to integrate a generic tensor-valued function $\boldsymbol{f} = \boldsymbol{f}(t, \boldsymbol{x})$, $\boldsymbol{f} : \mathbb{R} \times \mathbb{R}^3 \mapsto \mathcal{T}^n$, whose counter-domain is the n-th order tensor space,

$$\begin{gathered}\int\limits_{\Omega} \boldsymbol{f}\,\mathrm{d}v = \lim_{n\to\infty} \sum_{(\boldsymbol{x}_i, \Delta v_i)\in\mathfrak{S}(n,\Omega)} \boldsymbol{f}(\boldsymbol{x}_i)\,\Delta v_i = \lim_{n\to\infty} \left[\boldsymbol{f}(\boldsymbol{x}_1)\,\Delta v_1 + \boldsymbol{f}(\boldsymbol{x}_2)\,\Delta v_2 + \ldots\right]; \\ \mathfrak{S}(n,\Omega) = \left\{(\boldsymbol{x}_i, \Delta v_i)\,\forall i \in \{1, 2, \ldots, n\} \mid \boldsymbol{x}_i \in \Omega_i \subset \Omega,\ \ \Delta v_i = |\Omega_i|\right\}, \end{gathered} \tag{140}$$

where $\mathfrak{S}(n,\Omega)$ is the set of n partitions Ω_i of the domain Ω, and Δv_i is the volume of the i-th partition, Ω_i. As the integration operator has been rewritten as the sum of a set of infinite partitions of the domain, the differentiation operator can be applied directly using the product and chain rules,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\int\limits_{\Omega} \boldsymbol{f}\,\mathrm{d}v &= \lim_{n\to\infty} \sum_{(\boldsymbol{x}_i, \Delta v_i)\in\mathfrak{S}(n,\Omega)} \left[\frac{\partial \boldsymbol{f}}{\partial t}\Delta v_i + \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \cdot \frac{\partial \boldsymbol{x}_i}{\partial t}\Delta v_i + \boldsymbol{f}(\boldsymbol{x}_i)\,\frac{\partial \Delta v_i}{\partial t}\right] \\ = \lim_{n\to\infty} \sum_{(\boldsymbol{x}_i, \Delta v_i)\in\mathfrak{S}(n,\Omega)} \frac{\partial \boldsymbol{f}}{\partial t}\Delta v_i &+ \lim_{n\to\infty} \sum_{(\boldsymbol{x}_i, \Delta v_i)\in\mathfrak{S}(n,\Omega)} \left[\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \cdot \frac{\partial \boldsymbol{x}_i}{\partial t}\Delta v_i + \boldsymbol{f}(\boldsymbol{x}_i)\,\frac{\partial \Delta v_i}{\partial t}\right] \\ &= \int\limits_{\Omega} \frac{\partial \boldsymbol{f}}{\partial t}\,\mathrm{d}v + \underbrace{\lim_{n\to\infty} \sum_{(\boldsymbol{x}_i, \Delta v_i)\in\mathfrak{S}(n,\Omega)} \left[\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \cdot \frac{\partial \boldsymbol{x}_i}{\partial t}\Delta v_i + \boldsymbol{f}(\boldsymbol{x}_i)\,\frac{\partial \Delta v_i}{\partial t}\right]}_{\boldsymbol{\epsilon}}. \end{aligned} \tag{141}$$

As the domain Ω is a continuous collection of points defined by the position vector $\boldsymbol{x}$, Equation (141) enables us to prove that

$$\Omega \text{ is time-invariant and } \boldsymbol{f} \text{ is differentiable} \implies \frac{\mathrm{d}}{\mathrm{d}t}\int\limits_{\Omega} \boldsymbol{f}\,\mathrm{d}v = \int\limits_{\Omega} \frac{\partial \boldsymbol{f}}{\partial t}\,\mathrm{d}v. \tag{142}$$

Semantically, the differentiation and integration operators commute if the integration domain is invariant to time and the integrand is differentiable and smooth. This occurs because, given these conditions, the tensor $\boldsymbol{\epsilon}$ is null.

Returning to Eq. (139), it is now clear that the time derivative there **cannot** be carried into the integral, since the domain of the deformed configuration is a function of time. However, the mathematical construction of the reference configuration comes at hand for the very first time, since we can pull-back Eq. (139) to the material configuration and take the derivative there,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega} \boldsymbol{f}\,\mathrm{d}v = \frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_0} \boldsymbol{f}\left(\boldsymbol{x}\left(\boldsymbol{X}\right)\right) J\,\mathrm{d}V = \int_{\Omega_0} \frac{\partial \boldsymbol{f}}{\partial t} J + \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \cdot \frac{\partial \boldsymbol{x}}{\partial t} J + \boldsymbol{f}\frac{\partial J}{\partial t}\,\mathrm{d}V; \tag{143}$$

whose time derivative of the jacobian can be simplified using Eq. (106),

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega} \boldsymbol{f}\,\mathrm{d}v = \int_{\Omega_0} J\left[\frac{\partial \boldsymbol{f}}{\partial t} + \mathrm{grad}\left(\boldsymbol{f}\right)\cdot\boldsymbol{v}\right] + \boldsymbol{f}J\mathrm{div}\left(\boldsymbol{v}\right)\mathrm{d}V; \tag{144}$$

where $\boldsymbol{v}$ is the velocity vector of the particle attached to the position vector $\boldsymbol{X}$ in the reference configuration or to the position vector $\boldsymbol{x}$ in the deformed configuration. As the jacobian J is the determinant of the deformation gradient tensor, we rely on Eq. (72) to reduce Eq. (143) even further,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega} \boldsymbol{f}\,\mathrm{d}v = \int_{\Omega_0} \left[\frac{\partial \boldsymbol{f}}{\partial t} + \mathrm{grad}\left(\boldsymbol{f}\right)\cdot\boldsymbol{v} + \boldsymbol{f}\mathrm{div}\left(\boldsymbol{v}\right)\right] J\,\mathrm{d}V. \tag{145}$$

The integrand in Eq. (145) is a function with direct correspondence in the deformed configuration, which means that all quantities can be equally evaluated point-wise in the spatial configuration. Thus, the term $J\mathrm{d}V$ can be converted to the volume differential in the spatial configuration and, consequentially, the integral can be reevaluated in the deformed configuration,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega} \boldsymbol{f}\,\mathrm{d}v = \int_{\Omega} \left[\frac{\partial \boldsymbol{f}}{\partial t} + \mathrm{grad}\left(\boldsymbol{f}\right)\cdot\boldsymbol{v} + \boldsymbol{f}\mathrm{div}\left(\boldsymbol{v}\right)\right] \mathrm{d}v. \tag{146}$$

If function $\boldsymbol{f}$ is a scalar-valued function, using the product rule from Eq. (110), it is possible to compress Eq. (146) to

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega} f\,\mathrm{d}v = \int_{\Omega} \left[\frac{\partial f}{\partial t} + \mathrm{div}\left(f\boldsymbol{v}\right)\right] \mathrm{d}v. \tag{147}$$

Equation (146) and Equation (147) are the mathematical statement of the **Reynold's transport theorem**. This theorem systematically evaluates the time derivative of functions integrated in the spatial configuration.

Challenge 5.1

Show, using Einstein notation, that Eq. (146) can be rewritten using the product rule to

$$\frac{\mathrm{d}}{\mathrm{d}t} \int\limits_{\Omega} \boldsymbol{f} \,\mathrm{d}v = \int\limits_{\Omega} \left[\frac{\partial \boldsymbol{f}}{\partial t} + \mathrm{div}\left(\boldsymbol{f} \otimes \boldsymbol{v}\right) \right] \mathrm{d}v. \tag{148}$$

One particular case of the scalar version of Eq. (147) is when f is the spatial mass density function $\rho\,\left[\mathrm{kg/m^3}\right]$,

$$\frac{\mathrm{d}}{\mathrm{d}t} \int\limits_{\Omega} \rho \,\mathrm{d}v = \int\limits_{\Omega} \frac{\partial \rho}{\partial t} + \mathrm{grad}\left(\rho\right) \cdot \boldsymbol{v} + \rho \mathrm{div}\left(\boldsymbol{v}\right) \mathrm{d}v. \tag{149}$$

Note that the partial derivative of the density function plus its gradient projected onto the velocity vector field is the total derivative of the spatial density function (see Eq. (104)). Thereby, Equation (149) can be rewritten to

$$\frac{\mathrm{d}}{\mathrm{d}t} \int\limits_{\Omega} \rho \,\mathrm{d}v = \frac{\mathrm{d}m}{\mathrm{d}t} = \int\limits_{\Omega} \frac{\mathrm{d}\rho}{\mathrm{d}t} + \rho \mathrm{div}\left(\boldsymbol{v}\right) \mathrm{d}v, \tag{150}$$

which relates the time derivative of the density with the rate of change of mass in Ω. As we will discuss in Subsec. 5.4 and in Subsec. 5.5, Equation (150) can be simplified to a local form if the system is *closed*, i.e. the mass rate is zero. This local form is available at Eq. (160).

With the local relation for the density field in the case of a closed system, Equation (146) can be particularized for the case where the integrand is multiplied by the spatial density ρ,

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} \int\limits_{\Omega} \rho \boldsymbol{f} \,\mathrm{d}v &= \int\limits_{\Omega} \left[\frac{\partial}{\partial t}\left(\rho \boldsymbol{f}\right) + \mathrm{grad}\left(\rho \boldsymbol{f}\right) \cdot \boldsymbol{v} + \rho \boldsymbol{f} \mathrm{div}\left(\boldsymbol{v}\right) \right] \mathrm{d}v \\
= \int\limits_{\Omega} \frac{\mathrm{d}}{\mathrm{d}t}\left(\rho \boldsymbol{f}\right) + \rho \boldsymbol{f} \mathrm{div}\left(\boldsymbol{v}\right) \mathrm{d}v &= \int\limits_{\Omega} \frac{\mathrm{d}\rho}{\mathrm{d}t} \boldsymbol{f} + \rho \frac{\mathrm{d}\boldsymbol{f}}{\mathrm{d}t} + \rho \boldsymbol{f} \mathrm{div}\left(\boldsymbol{v}\right) \mathrm{d}v \\
&= \int\limits_{\Omega} \rho \frac{\mathrm{d}\boldsymbol{f}}{\mathrm{d}t} + \left[\frac{\mathrm{d}\rho}{\mathrm{d}t} + \rho \mathrm{div}\left(\boldsymbol{v}\right) \right] \boldsymbol{f} \,\mathrm{d}v.
\end{aligned} \tag{151}$$

The term between brackets in the last line of Eq. (151) is precisely the relation for mass conservation, Eq. (160). If the system conserves mass, as most systems do, Equation (151) simplifies to

$$\frac{\mathrm{d}}{\mathrm{d}t}\int\limits_{\Omega} \rho \boldsymbol{f}\,\mathrm{d}v = \int\limits_{\Omega} \rho \frac{\mathrm{d}\boldsymbol{f}}{\mathrm{d}t}\,\mathrm{d}v. \tag{152}$$

5.4. Global to local conservation equations

Suppose that we pick a point in the spatial configuration, which is located by the position vector $\boldsymbol{x}_c$. Let this point be the centroid of an arbitrary region in the domain Ω_i that completely involves such point, $\boldsymbol{x}_i \in \Omega_i \subset \Omega$. This region has a boundary Γ_i that has no intersection with the boundary Γ of the domain Ω, i.e. $\Gamma_i \cap \Gamma = \emptyset$. We might particularize the generic balance equation from Eq. (137) to the subregion Ω_i,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int\limits_{\Omega_i} \boldsymbol{c}\,\mathrm{d}v = \int\limits_{\Omega_i} \boldsymbol{q} - \mathrm{div}\,(\boldsymbol{a})\,\mathrm{d}v. \tag{153}$$

Applying the Reynold's transport theorem (see Eq. (146)) and reorganizing the terms, Eq. (153) can be rewritten to

$$\int\limits_{\Omega_i} \frac{\partial \boldsymbol{c}}{\partial t} + \mathrm{grad}\,(\boldsymbol{c}) \cdot \boldsymbol{v} + \boldsymbol{c}\mathrm{div}\,(\boldsymbol{v}) + \mathrm{div}\,(\boldsymbol{a}) - \boldsymbol{q}\,\mathrm{d}v = \boldsymbol{0}. \tag{154}$$

Equation (154) must be satisfied to any arbitrary region in space. We suppose with no loss of physical significance that the integrand in Eq. (154) is smooth and differentiable. Imagine that this region becomes progressively smaller and closer to the point $\boldsymbol{x}_i$, as the integrand is smooth, the latter will become ever closer to a linear function and the result of the integration will asymptotically converge to the mean value of that plane (the value of the integrand at $\boldsymbol{x}_i$) times the volume of the subregion—which comes from the mean value theorem for integrals. Thus, if this result must be null, the value of the function at this point must be zero. As this must happen for any arbitrary point $\boldsymbol{x}_i$, the integrand must be identically zero, which summarizes to

$$\frac{\partial \boldsymbol{c}}{\partial t} + \mathrm{grad}\,(\boldsymbol{c}) \cdot \boldsymbol{v} + \boldsymbol{c}\mathrm{div}\,(\boldsymbol{v}) + \mathrm{div}\,(\boldsymbol{a}) - \boldsymbol{q} = \boldsymbol{0} \ \ \forall \boldsymbol{x} \in \Omega. \tag{155}$$

Equation (155) is the mathematical statement of *local* conservation law, since it is valid to every point in the domain. The local form is the basis for all partial differential equations arising in continuum mechanics from balance principles. The local and integral forms of conservation laws are both physically and mathematically complete and correct; their usage on the other hand will be tightly linked to the method we will use to numerically find solutions of the fields involved in these problems, such as displacement, velocity, pressure, etc. For the method of finite differences, for instance, the local form is straightforward to be used; for the finite element method, integral equations are more appropriate.

5.5. Conservation of mass

We will not follow the historical order to which such conservation laws had been proposed. We will prioritize clarity and an order of increasing conceptual complexity. This way, we start by conservation of mass, which was proposed by famous chemist Lavoisier in the 18$^{\text{th}}$ century. In modern notation we postulate that *all matter leaving a body by its boundaries must balance all mass fluxes coming into the body and the generation of mass in it.* Thus, we shall propose the equation

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \int\limits_{\Omega} \dot{m}_g \,\mathrm{d}v - \int\limits_{\Gamma} \dot{\boldsymbol{m}}_o \cdot \boldsymbol{n} \,\mathrm{d}s, \tag{156}$$

where m is the mass contained in the domain Ω at any instance t ([kg]); $\dot{m}_g$ is the mass generation rate per volume $\left(\left[\mathrm{kg/sm^3}\right]\right)$ in the domain Ω, and $\dot{\boldsymbol{m}}_o$ is the outward rate flux vector of mass $\left(\left[\mathrm{kg/sm^2}\right]\right)$.

Using the divergence theorem (see Eq. (107), Equation (156) can be rewritten to an integration over the domain only,

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \int\limits_{\Omega} \dot{m}_g - \mathrm{div}\left(\dot{\boldsymbol{m}}_o\right) \mathrm{d}v. \tag{157}$$

In addition to that, the mass of the domain can be interpreted as the integration of a volumetric density of mass ρ (the conventional density one usually thinks of). Plugging this definition with the Reynold's transport theorem into Eq. (157) yields

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} \int\limits_{\Omega} \rho \,\mathrm{d}v = \int\limits_{\Omega} \frac{\partial \rho}{\partial t} + \mathrm{div}\left(\rho \boldsymbol{v}\right) \mathrm{d}v = \int\limits_{\Omega} \dot{m}_g - \mathrm{div}\left(\dot{\boldsymbol{m}}_o\right) \mathrm{d}v \\ \implies \int\limits_{\Omega} \frac{\partial \rho}{\partial t} + \mathrm{div}\left(\rho \boldsymbol{v} + \dot{\boldsymbol{m}}_o\right) - \dot{m}_g \,\mathrm{d}v = 0. \end{aligned} \tag{158}$$

If the system is closed and there is no generation nor mass transfer through the boundary, Equation (158) becomes the usually seen integral version of mass conservation,

$$\int\limits_{\Omega} \frac{\partial \rho}{\partial t} + \mathrm{div}\left(\rho \boldsymbol{v}\right) \mathrm{d}v = 0; \tag{159}$$

which, by using the principle derived in Subsec. 5.4, develops into the local form of the mass conservation equation,

$$\frac{\partial \rho}{\partial t} + \mathrm{div}\left(\rho \boldsymbol{v}\right) = 0. \tag{160}$$

5.6. Conservation of momentum

Newton derived two momenta—the linear, which accounts for linear motion, and angular, which account for kinematics analogue to rotations. Newton postulated that the laws of motions, which regard displacements, velocities, and accelerations, are governed by the conservation in time of this quantity. To this end, the illustrious English mathematician proposed that the rate of change of momentum is equal to the sum of forces acting on the body. In classical continuum mechanics, point forces of the mechanics derived by Newton become fields over the domain and over the boundary. These two categories of force will play the exact same role as the generic generation volume density and the net flux on the boundary, as in Eq. (138).

The net source term in Eq. (136) is the sum of all forces spread across the domain as a field of a force volumetric density,

$$\bar{\boldsymbol{q}} = \int_{\Omega} \boldsymbol{b} \, \mathrm{d}v, \quad \boldsymbol{b} \in \mathbb{R}^3, \quad \boldsymbol{b}: \, \left[\mathrm{N}/\mathrm{m}^3\right]. \tag{161}$$

The term related to the net flux on the boundary (see Eq. (135)) is the sum of forces (surface area density) on the boundary. Note that such force area density is exactly the definition of traction that we presented in Sec. 4. We will, however, change the sign of the flux integral to make it consistent with Eq. (137), since the traction on the boundary adds up to the rate of change of linear momentum (all forces are summed, no matter if they are sources or fluxes),

$$\bar{\boldsymbol{a}} = -\int_{\Gamma} \boldsymbol{t} \, \mathrm{d}s = -\int_{\Gamma} \boldsymbol{\sigma} \boldsymbol{n} \, \mathrm{d}s = -\int_{\Omega} \mathrm{div}\left(\boldsymbol{\sigma}\right) \mathrm{d}v. \tag{162}$$

The third and last term, the term related to the time derivative of a quantity, is the rate of change of linear momentum. Linear momentum can be easily extended from Newton classical mechanics of particles and rigid bodies using a mass differential,

$$\frac{\mathrm{d}\boldsymbol{L}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega} \mathrm{d}\boldsymbol{L} = \frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega} \boldsymbol{v} \, \mathrm{d}m = \frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega} \rho \boldsymbol{v} \, \mathrm{d}v. \tag{163}$$

Considering a system that conserves mass and joining the three pieces together, one gets the integral form for the conservation of linear momentum,

$$\int_{\Omega} \rho \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} - \boldsymbol{b} - \mathrm{div}\left(\boldsymbol{\sigma}\right) \mathrm{d}v = \boldsymbol{0}. \tag{164}$$

Then, using the rationale from Subsec. 5.4, the local form is

$$\operatorname{div}(\boldsymbol{\sigma}) + \boldsymbol{b} = \rho \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t}. \tag{165}$$

The remaining momentum is the angular one, which balances moments (forces multiplied by a lever direction from a fixed point). Thus, we start by evaluating the moments generated by the body forces, which will be the source of moments from a force field

$$\bar{\boldsymbol{q}} = \int_{\Omega} \boldsymbol{r} \times \boldsymbol{b}\, \mathrm{d}v, \tag{166}$$

where $\boldsymbol{r}$ is the lever direction from a fixed point in space to the point where the body force is calculated. The net flux of moment is computed by the sum of contributions of the moments generated by the tractions in the boundary. Again, the sign will be negative to make the generic balance equation sum all contributions of moments with a positive sign. The resulting net moment generated by tractions on the boundary surface is

$$\bar{\boldsymbol{a}} = -\int_{\Gamma} \boldsymbol{r} \times \boldsymbol{t}\, \mathrm{d}s = \int_{\Gamma} \boldsymbol{r} \times \boldsymbol{\sigma}\boldsymbol{n}\, \mathrm{d}s, \tag{167}$$

Using Eq. (112), the divergence theorem transforms the surface integral to the volume integral

$$\bar{\boldsymbol{a}} = -\int_{\Omega} \boldsymbol{\varepsilon} : \boldsymbol{\sigma}^{\mathrm{T}} + \boldsymbol{r} \times \operatorname{div}(\boldsymbol{\sigma})\, \mathrm{d}v = -\int_{\Omega} \operatorname{axl}\left(\boldsymbol{\sigma}^{\mathrm{T}}\right) + \boldsymbol{r} \times \operatorname{div}(\boldsymbol{\sigma})\, \mathrm{d}v, \tag{168}$$

where $\boldsymbol{\varepsilon}$ is the third-order Levi-Civita permuation tensor.

The derivative of the sum of the contribution of angular moment of all particles with respect to a fixed reference point is given as

$$\begin{aligned}\frac{\mathrm{d}\boldsymbol{J}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega} \boldsymbol{r} \times \rho\boldsymbol{v}\, \mathrm{d}v = \boldsymbol{\varepsilon} : \left[\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega} \rho\boldsymbol{r} \otimes \boldsymbol{v}\, \mathrm{d}v\right] &= \boldsymbol{\varepsilon} : \int_{\Omega} \rho \frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{r} \otimes \boldsymbol{v})\, \mathrm{d}v \\ &= \boldsymbol{\varepsilon} : \int_{\Omega} \rho \left[\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} \otimes \boldsymbol{v} + \boldsymbol{r} \otimes \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t}\right] \mathrm{d}v.\end{aligned} \tag{169}$$

The lever vector from the fixed reference point, $\boldsymbol{x}_0$, to any particle is represented by the expression $\boldsymbol{r} = \boldsymbol{x} - \boldsymbol{x}_0$. Thus, the total derivative of the lever vector is the velocity field vector, $\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \boldsymbol{v}$. As we add this expression to Eq. (169), it yields

$$\frac{\mathrm{d}\boldsymbol{J}}{\mathrm{d}t} = \boldsymbol{\varepsilon} : \left[\int_{\Omega} \rho \boldsymbol{v} \otimes \boldsymbol{v} \, \mathrm{d}v + \int_{\Omega} \rho \boldsymbol{r} \otimes \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} \, \mathrm{d}v \right] = \int_{\Omega} \rho \boldsymbol{v} \times \boldsymbol{v} + \rho \boldsymbol{r} \times \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} \, \mathrm{d}v. \tag{170}$$

The vector product between a vector and itself is always the null vector, hence the term $\boldsymbol{v} \times \boldsymbol{v} = \boldsymbol{0}$. Combining Eq. (166), Eq. (168), and Eq. (170) to the generic balance equation in Eq. (138) produces the equation of conservation of angular momentum,

$$\begin{aligned} \int_{\Omega} \rho \boldsymbol{r} \times \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} \, \mathrm{d}v &= \int_{\Omega} \boldsymbol{r} \times \boldsymbol{b} \, \mathrm{d}v + \int_{\Omega} \mathrm{axl} \left(\boldsymbol{\sigma}^{\mathrm{T}} \right) + \boldsymbol{r} \times \mathrm{div} \left(\boldsymbol{\sigma} \right) \mathrm{d}v \\ \implies \int_{\Omega} \rho \boldsymbol{r} &\times \left[\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} - \boldsymbol{b} - \mathrm{div} \left(\boldsymbol{\sigma} \right) \right] - \mathrm{axl} \left(\boldsymbol{\sigma}^{\mathrm{T}} \right) \mathrm{d}v = \boldsymbol{0}. \end{aligned} \tag{171}$$

Conversely, as discussed in Subsec. 5.4, the integral form of Eq. (171) can be transmuted to a local form,

$$\rho \boldsymbol{r} \times \left[\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} - \boldsymbol{b} - \mathrm{div} \left(\boldsymbol{\sigma} \right) \right] - \mathrm{axl} \left(\boldsymbol{\sigma}^{\mathrm{T}} \right) = \boldsymbol{0}. \tag{172}$$

Observe that the term in brackets in Eq. (172) is identically the local form of the balance of linear momentum. Therefore, the inside of the brackets is null. We are left with the axial vector of the transpose of the Cauchy stress tensor,

$$\mathrm{axl} \left(\boldsymbol{\sigma}^{\mathrm{T}} \right) = \boldsymbol{0} \implies \boldsymbol{\sigma}^{\mathrm{T}} \in \mathrm{Sym}^3 \implies \boldsymbol{\sigma} \in \mathrm{Sym}^3. \tag{173}$$

In plain words, we could demonstrate that for classical formulation of continuum mechanics, where there is no body moments nor moments in the boundary, the balance of angular momentum is satisfied by the balance of linear momentum and, consequently, the Cauchy stress tensor must be symmetric.

5.7. Conservation of power

Energy and power are mathematical constructions that simplify the analysis of complex problems by recasting them into a potential or a scalar field. Such potential field stores an abstract quantity, known as *energy* or *work*, that is capable of transforming physical objects by all sorts of manners—moving, changing shape, heating, glowing, magnetizing, etc. The rate in time to which energy changes is known as power. Energy (and its time rate by consequence) is, if not the most, one of the most used physical quantities that no one actually knows what it really is. That is because we know and measure (or conceptualize) it indirectly by the aforementioned effects on matter.

Treating energy and power as the abstractions they really are, we could frame them as *functionals* (functions that receive other functions as arguments and return a real number, the inner product of two vector fields is an example of a functional) of the kinematics fields of the observed phenomenon, such as the displacement field. Such functionals have been the focus of the so called *energy-based* formulations, as exemplified by Lagrangian and Hamiltonian mechanics. We will refer to the work of Ciarlet [9] on the basis of energy and potentials in continuum mechanics, then to Ciarlet et al. [10], where the concept of duality emerges from function spaces and their dual spaces in extrema problems (minimum, maximum, or saddle point).

As we presented in Subsec. 5.2 and, subsequently discussed in Subsec. 5.4, a myriad of conservation laws branch out to partial differential equations (PDEs) (the local form of the balance equations). These PDEs possess a divergent structure, due to the divergence operator arising from the net flux. The divergent structure enables the use of an inner product of the PDE with a function that is a generic variation of the field that is the solution to the PDE. This approach is the origin of methods of virtual work and virtual power [9, 11]. Thus, one can conclude that energy can be postulated, as we will do in this introductory text, or as a consequence of a variational principle over PDEs.

There are many forms of energy: thermal, heat, chemical, mechanical, kinetic, etc. We opted to postulate their equations as this is the approach one is used to since basic courses in physics. Moreover, the variational approach would require a background in functional analysis, which is not required at this point. We also opted to present power instead of energy, since power will be readily used to derive the second law of thermodynamics. In the next subsubsections, we will present different forms of power in different contexts.

We divide the presented forms of power into three broad categories: *storage*, *source*, and *transport*. The storage-related forms of energy, as the name already suggests, store energy to be used afterward; storage-related forms of power are simply the time rate of the same forms of energy. The source-related forms of power are generated in the domain due to sources. The transport-related forms of power are time rates in their own right. Transport-related forms of power are linked to energy coming in and out of the domain, through the boundary as fluxes. We show a limited set of powers, since we will focus on thermomechanical problems. The reader, if interested, can look for specialized literature in power forms related to other phenomena, such as electromagnetism to name one example.

In contrast to storage-related energies, the source and transport-related powers are postulated as power; in other words, the energy does not need to be proposed and, then, differentiated with respect to time. This can be fully discussed and justified using the variational approach, such as in [10, 11].

5.7.1. Storage-related forms of power

We define two forms of storage-related energy, internal energy and kinetic energy. Both of them are defined as contained by individual particles, which have infinitesimal mass. Thus, using the mass density function, it is possible to integrate the contribution of each particle to the total amount of energy. Storage-related forms of energy cannot be negative, they must be absolute values. However, their time rates can indeed be negative.

Kinetic energy is the energy associated with the movement of a particle. Thus, the kinetic energy is given by the inner product of the velocity vector with itself. The kinetic energy of the whole domain is defined as

$$\mathcal{K} = \int \frac{1}{2}\boldsymbol{v} \cdot \boldsymbol{v} \, \mathrm{d}m = \frac{1}{2} \int_{\Omega} \rho \boldsymbol{v} \cdot \boldsymbol{v} \, \mathrm{d}v. \tag{174}$$

Note that, due to the property of positive-definiteness of the inner product (see Eq. (60)), the kinetic energy, as defined, is always positive.

The internal energy, on the other hand, does not explicitly depend upon other physical quantities, it is rather left as a field in its own right. Internal energy is a bit exotic to fully comprehend and is a generic construction,

$$\mathcal{E} = \int e \, \mathrm{d}m = \int_{\Omega} \rho e \, \mathrm{d}v; \tag{175}$$

where $e : \mathbb{R}^3 \mapsto \mathbb{R}$ is a field of internal energy density.

We suppose that the physical phenomena are posed in a closed system with respect to mass exchange, such that mass is conserved. Hence, we can evaluate the time rate of both storage-related energies using the density-based version of the Reynold's transport theorem, Eq. (152),

$$\frac{\mathrm{d}\mathcal{K}}{\mathrm{d}t} = \dot{\mathcal{K}} = \frac{1}{2} \int_{\Omega} \rho \left[\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} \cdot \boldsymbol{v} + \boldsymbol{v} \cdot \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} \right] \mathrm{d}v = \int_{\Omega} \rho \boldsymbol{v} \cdot \dot{\boldsymbol{v}} \, \mathrm{d}v; \tag{176}$$

and for the time rate of the internal energy,

$$\frac{\mathrm{d}\mathcal{E}}{\mathrm{d}t} = \dot{\mathcal{E}} = \int_{\Omega} \rho \dot{e} \, \mathrm{d}v. \tag{177}$$

5.7.2. Source-related forms of power

We postulate the power supplied by heat sources, i.e. sources of flux of thermal energy, which naturally appear in problems of thermoelasticity,

$$\mathcal{Q}_s = \int \mathrm{d}\mathcal{Q}_s = \int_{\Omega} r \, \mathrm{d}v, \tag{178}$$

where $r : \mathbb{R}^3 \mapsto \mathbb{R}$ is a field of heat generation per volume $\left[\mathrm{J/sm^3}\right]$.

The field of body forces, $\boldsymbol{b}$, introduced in Subsec. 5.6 also add power to the system, and their *power conjugate* is the velocity field, which means that the inner product of the body forces with the velocity field result in power. This statement is summarized mathematically as

$$\mathcal{P}_s = \int_{\Omega} \boldsymbol{b} \cdot \boldsymbol{v} \, \mathrm{d}v. \tag{179}$$

5.7.3. Transport-related forms of power

In Subsec. 5.1, we discussed fluxes of quantities *leaving* the domain through its boundary. In the context of power, it is not different. Thus, we will define the following equations as power **leaving** the body. The first outward flux of energy is heat flux, which is given by

$$\mathcal{Q}_t = \int_{\Gamma} \boldsymbol{q} \cdot \boldsymbol{n} \, \mathrm{d}s = \int_{\Omega} \mathrm{div}\,(\boldsymbol{q}) \, \mathrm{d}v, \tag{180}$$

where $\boldsymbol{q}$ is the outward heat flux defined on the boundary Γ $\left[\mathrm{J/sm^2}\right]$.

As do the body forces, the traction field on the boundary is also power conjugate with the velocity field. This power, however, is added to the system instead of taken from it; thus, we will write the power introduced by tractions with a negative sign (the same reasons for the negative sign in Eq. (162)),

$$\mathcal{P}_t = -\int_{\Gamma} \boldsymbol{t} \cdot \boldsymbol{v} \, \mathrm{d}s = -\int_{\Gamma} \boldsymbol{v} \cdot \boldsymbol{t} \, \mathrm{d}s = -\int_{\Gamma} \boldsymbol{v} \cdot \boldsymbol{\sigma} \boldsymbol{n} \, \mathrm{d}s = -\int_{\Gamma} \left(\boldsymbol{\sigma}^{\mathrm{T}} \boldsymbol{v}\right) \cdot \boldsymbol{n} \, \mathrm{d}s = -\int_{\Omega} \mathrm{div}\left(\boldsymbol{\sigma}^{\mathrm{T}} \boldsymbol{v}\right) \mathrm{d}v. \tag{181}$$

5.7.4. First law of thermodynamics

The first law of thermodynamics postulates that energy is conserved in a closed system. We are interested in thermomechanical problems; hence, with the concluded derivation of the equations of powers involved, we can use the generic balance equation derived in Subsec. 5.2 to yield the integral form of the first law of thermodynamics. To this end, we will consider the sum of the storage-related energies as the physical quantity whose rate must be calculated (see $\bar{\boldsymbol{c}}$ in Eq. (138)), and the source and flux terms will be added to the right-hand side,

$$\frac{\mathrm{d}}{\mathrm{d}t} \left[\mathcal{K} + \mathcal{E}\right] = \mathcal{Q}_s + \mathcal{P}_s - \mathcal{Q}_t - \mathcal{P}_t. \tag{182}$$

The power relations we derived in Eq. (176), Eq. (177), Eq. (178), Eq. (179), Eq. (180), and Eq. (181) can be plugged into Eq. (182) to yield a meaningful version of the first law of thermodynamics for thermomechanical problems,

$$\int_{\Omega} \rho \left[\dot{e} + \boldsymbol{v} \cdot \dot{\boldsymbol{v}}\right] \mathrm{d}v = \int_{\Omega} r + \boldsymbol{b} \cdot \boldsymbol{v} - \operatorname{div}(\boldsymbol{q}) + \operatorname{div}\left(\boldsymbol{\sigma}^{\mathrm{T}} \boldsymbol{v}\right) \mathrm{d}v. \tag{183}$$

Using the product rule for the divergence of the Cauchy stress tensor multiplied by the velocity vector (see Eq. (111)), Equation (183) can be rearranged to

$$\int_{\Omega} \rho \dot{e} \,\mathrm{d}v = \int_{\Omega} r - \operatorname{div}(\boldsymbol{q}) + \left[\operatorname{div}(\boldsymbol{\sigma}) + \boldsymbol{b} - \rho \dot{\boldsymbol{v}}\right] \cdot \boldsymbol{v} + \boldsymbol{\sigma} : \operatorname{grad}(\boldsymbol{v}) \,\mathrm{d}v. \tag{184}$$

The term inside brackets is precisely the local form of the balance equation of linear momentum (see Eq. (165)). Consequently, Equation (184) can be once again simplified to

$$\int_{\Omega} \rho \dot{e} + \operatorname{div}(\boldsymbol{q}) - r - \boldsymbol{\sigma} : \operatorname{grad}(\boldsymbol{v}) \,\mathrm{d}v = 0; \tag{185}$$

conversely, the equivalent local form (see Subsec. 5.4) is

$$\rho \dot{e} + \operatorname{div}(\boldsymbol{q}) - r - \boldsymbol{\sigma} : \operatorname{grad}(\boldsymbol{v}) = 0. \tag{186}$$

We discussed how the velocity field is the power conjugate of the field of body forces and the traction on the boundary. Observe in Eq. (185) and in Eq. (186) that the Cauchy stress tensor is taken, alongside the spatial velocity gradient, by the inner product defined by the double contraction. For this reason, the Cauchy stress tensor generates power with the spatial gradient of velocity and, thus, the Cauchy stress tensor and the spatial gradient of velocity are power conjugates as well. Furthermore, as the Cauchy stress tensor yields the internal tractions at any point of the domain given any normal vector, the term $\boldsymbol{\sigma} : \operatorname{grad}(\boldsymbol{v})$ is known as the *internal mechanical power*.

We proved that the Cauchy stress tensor is symmetric in classical continuum theory, for there are no body moments nor boundary moments. The double contraction of a symmetric tensor with a skew-symmetric tensor is null, hence the Cauchy stress tensor contracts only with the symmetric part of the spatial gradient of velocity. Therefore, Equation (186) can be rewritten to

$$\rho \dot{e} + \operatorname{div}(\boldsymbol{q}) - r - \boldsymbol{\sigma} : \operatorname{sym}\big(\operatorname{grad}(\boldsymbol{v})\big) = \rho \dot{e} + \operatorname{div}(\boldsymbol{q}) - r - \boldsymbol{\sigma} : \boldsymbol{d} = 0, \tag{187}$$

where $\boldsymbol{d}$ is the *symmetric spatial gradient of velocity.* Following the rationale stated before, $\boldsymbol{\sigma}$ and $\boldsymbol{d}$ are also power conjugates.

5.8. Brief concluding commentary

In this section, we defined how the body interacts with the exterior with the intermediate of fluxes on its boundary; then, we defined how sources generate point-wise quantities as fields. With those definitions, we postulated a generic conservation law and its balance equation in integral and local forms. Finally, the Reynold's transport theorem satisfied our demand for time rates.

The most generic (and simple) principle we stated, then, enabled us to derive the most useful PDEs and their integral forms. To crown our efforts with green laurels, we were able even to deduce the first law of thermodynamics, a cornerstone of physics, from that basic and generic principle.

Exercise 5.1
Derive again all balance equations deduced before for the presented conservation laws, but in material configuration.

6. Second law of thermodynamics

In Subsec. 5.7, we derived the first law of thermodynamics as a consequence of the conservation of power using the generic pattern for conservation laws. The first law of thermodynamics states that such physical quantity called energy must be conserved; thus, it must be converted from one mode to another—mechanical to thermal, electromagnetic to mechanical, etc. Each one of those conversions between modes of energy is called a *process*.

The first law of thermodynamics, however, does not state if there is a particular *order* in each process; if they happen in a particular sequence or if they are *reversible*. Take an example: we observe that heat seems to flow from a hotter region to a colder region in space, there seems to be a preferable direction; in addition to that, once heat has flowed to the once colder region, it does not appear to flow back spontaneously. With this example, one reasonably understands that there must be an order and that spontaneous reversibility is not always the case.

Given the structure and framework of the scientific method, some important and thorny questions remain: is it possible to develop a scientific theory around those ideas? Can we prove that processes follow such an order? Is our observation of the heat-transfer example a concluding evidence to the hypotheses of order and lack of reversibility? If we run the heat-transfer experiment a thousand times and perceive that heat flows from the hotter to the colder regions, is it sufficient to assert that it will happen again at the 1001$^{\text{st}}$ experiment? These questions plagued the recently-born branch of physics, thermodynamics, at the dawn of the 19$^{\text{th}}$ century.

An Austrian mathematician and physicist, Ludwig Boltzmann, came with a solution and established a new framework inside thermodynamics, which has been called *statistical mechanics*. Boltzmann looked into the backbone of matter—molecules, a concept that was neither fully understood nor fully accepted by the scientific community in the early 19^{th} century. Boltzmann contributed to the understanding that the state of agitation of molecules encapsulated our abstract concept of energy, and that their behavior would shape macroscopic observations of matter and its interactions.

Ludwig Boltzmann postulated that the macroscopic observation of thermodynamic processes must be statistical, since the number of molecules in a system is astronomically high to construct a deterministic framework upon it. As a consequence, Boltzmann derived that our construction and observations of preferable directions and irreversibility (the lack of spontaneous reversibility) in thermodynamic processes is merely the result of overwhelming odds, in such probability distributions, to the said directions. Finally, Boltzmann took the concept of *entropy* from Clausius to consolidate irreversibility and to pose a macroscopic quantization of the probabilistic view on molecular mechanics.

Entropy is one of the most misunderstood concepts in the whole of physics. It is often interpreted as the *degree of disorder* in a system. That is not a formal understanding, since it is more of a concept to summarize the probabilistic framework laying underneath and to determine the *arrow of time* (why time always *flow* from the past to the future due to irreversible processes). Entropy can be transported in the sense that it can be traded in and out of the body to its surroundings. Although, it is postulated that its production within a body cannot be negative.

Returning to our example, heat spontaneously flows from the hotter to the colder regions. However, heat can flow from a colder to a hotter region, thus lowering the entropy of the colder region, but this must take additional power to happen. The input of power increases the entropy of the system, which asserts that the net production of entropy is not negative. Consequently, going against the preferable direction of an irreversible process costs energy to produce additional net entropy; which, in turn, translates to a perceived sense of time and unequivocal order in thermodynamical processes.

We will see in this section that the second law of thermodynamics is fundamental to defining constitutive models, such as the Fourier law of heat transfer and the relation between stress and strain tensors.

6.1. Clausius-Duhem inequality

Entropy can be generated through sources and transported to the surrounding boundary through net fluxes. The net balance of entropy can also be tracked as a field in the domain. Thus, entropy can be evaluated in a balance equation just like that of Eq. (137). For this reason, we shall define the individual terms of the balance equation first.

The net entropy of a body will be denoted as the integration on the domain of a entropy mass density, η $\left[\text{J/kgK}\right]$, where [K] is kelvin, the absolute temperature unit. Thus, the net entropy time rate is evaluated using the Reynold's transport theorem for quantities multiplied by the mass density field, Eq. (152),

$$\frac{\mathrm{d}}{\mathrm{d}t}\underbrace{\int\limits_{\Omega}\eta\,\mathrm{d}m}_{\text{net entropy of }\Omega} = \frac{\mathrm{d}}{\mathrm{d}t}\int\limits_{\Omega}\eta\rho\,\mathrm{d}v \xrightarrow[\text{theorem}]{\text{Reynold's}} \frac{\mathrm{d}}{\mathrm{d}t}\int\limits_{\Omega}\rho\eta\,\mathrm{d}v = \int\limits_{\Omega}\rho\frac{\mathrm{d}\eta}{\mathrm{d}t}\,\mathrm{d}v = \int\limits_{\Omega}\rho\dot{\eta}\,\mathrm{d}v. \tag{188}$$

Clausius defined entropy as a quotient of heat flux by absolute temperature. This definition will be used to establish entropy transport and generation for the classical format of the Clausius-Duhem inequality. However, it is important to note that this definition is not consistent with the kinetic theory of gases derived by Boltzmann [1]. This relation is valid only when all possible small chunks of the body are in local *thermodynamic equilibrium* [12], i.e. there is no local transfer of heat or temperature gradients. This assumption yields small deviations to the kinetic theory once the temperature gradients are small. For this reason, this assumption will be used in continuum mechanics to derive the classic form of the Clausius-Duhem inequality.

Using the assumption of thermodynamic equilibrium, we propose the generation of entropy (source) as the quotient of the field generation of heat, $r : \mathbb{R}^3 \mapsto \mathbb{R}$, by the absolute temperature field, $T : \mathbb{R}^3 \mapsto \mathbb{R}^+$,

$$\bar{\boldsymbol{q}} = \int\limits_{\Omega}\frac{r}{T}\,\mathrm{d}v. \tag{189}$$

Conversely, the same postulate on entropy can be used to derive the entropy flux through the boundary. The transport of entropy on the boundary, hence, is evaluated by the ratio of heat flux on the boundary by the absolute temperature,

$$\bar{a} = \int\limits_{\Gamma}\frac{1}{T}\boldsymbol{q}\cdot\boldsymbol{n}\,\mathrm{d}s = \int\limits_{\Omega}\operatorname{div}\left(\frac{\boldsymbol{q}}{T}\right)\mathrm{d}v. \tag{190}$$

The Clausius-Duhem inequality states, in words, that the time rate of net entropy in the body must be equal or greater than the balance of entropy generation due to heat sources and the net flux of entropy (transport). This can be exactly framed by Eq. (137) if the equality sign should be modified by the "greater than" symbol. Using the product rule for the divergence theorem (see Eq. (110)), the Clausius-Duhem inequality becomes

$$\begin{aligned}\int\limits_{\Omega}\rho\dot{\eta}\,\mathrm{d}v \geq \int\limits_{\Omega}\frac{r}{T} - \operatorname{div}\left(\frac{\boldsymbol{q}}{T}\right)\mathrm{d}v &= \int\limits_{\Omega}\frac{r}{T} - \frac{1}{T}\operatorname{div}(\boldsymbol{q}) - \operatorname{grad}\left(\frac{1}{T}\right)\cdot\boldsymbol{q}\,\mathrm{d}v \\ &= \int\limits_{\Omega}\frac{1}{T}\big[r - \operatorname{div}(\boldsymbol{q})\big] + \frac{1}{T^2}\operatorname{grad}(T)\cdot\boldsymbol{q}\,\mathrm{d}v.\end{aligned} \tag{191}$$

Equation (191) can be rewritten to show the characteristic non-negativity of the net entropy generation, as discussed previously. Besides, the term in brackets can be substituted according to the local form of the first law of thermodynamics, see Eq. (186),

$$\int_{\Omega} \rho\dot{\eta} + \frac{1}{T}\left[\boldsymbol{\sigma} : \mathrm{grad}\,(\boldsymbol{v}) - \rho\dot{e} - \frac{1}{T}\mathrm{grad}\,(T) \cdot \boldsymbol{q}\right] \mathrm{d}v \geq 0; \tag{192}$$

or

$$\int_{\Omega} \rho\left[\dot{\eta} - \frac{\dot{e}}{T}\right] + \frac{1}{T}\boldsymbol{\sigma} : \mathrm{grad}\,(\boldsymbol{v}) - \frac{1}{T^2}\mathrm{grad}\,(T) \cdot \boldsymbol{q}\,\mathrm{d}v \geq 0. \tag{193}$$

To once again simplify this inequality, the Duhamel law of conduction can be inserted, $\boldsymbol{q} = -\boldsymbol{k}\mathrm{grad}\,(T)$, where $\boldsymbol{k}$ is the second-order conductivity tensor (Duhamel law of conduction is a generalization of the Fourier law of conduction to anisotropic conduction of heat). The Fourier law states that conductive heat transfer occurs in the opposite direction of the temperature gradient. Hence, the Fourier and Duhamel laws are **constitutive** models linking temperature gradient to heat flux. Substituting the Duhamel law into Eq. (193) yields the classical integral form of the Clausius-Duhem inequality,

$$\int_{\Omega} \rho\left[\dot{\eta} - \frac{\dot{e}}{T}\right] + \frac{1}{T}\boldsymbol{\sigma} : \mathrm{grad}\,(\boldsymbol{v}) + \frac{1}{T^2}\mathrm{grad}\,(T) \cdot \boldsymbol{k}\mathrm{grad}\,(T)\,\mathrm{d}v \geq 0; \tag{194}$$

whose local form, following the arguments of Subsec. 5.4 and multiplying both sides by the absolute temperature, becomes

$$\rho\left[T\dot{\eta} - \dot{e}\right] + \boldsymbol{\sigma} : \mathrm{grad}\,(\boldsymbol{v}) + \frac{1}{T}\mathrm{grad}\,(T) \cdot \boldsymbol{k}\mathrm{grad}\,(T) \geq 0. \tag{195}$$

Once net entropy is generated, i.e Equation (195) is strictly greater than 0, energy is dissipated. Dissipation of energy often happens due to its conversion to thermal energy, since part of that energy, converted to an increase in temperature, cannot be converted back into useful work. For this reason, a new metric of *available* or *free* energy must be introduced. Using the Legendre-Fenchel transform, the *Helmholtz free-energy density* (per unit reference volume) is introduced, which is defined as

$$\psi = \rho_0\left[e - T\eta\right] \quad \left[\mathrm{J/m^3}\right] \implies \dot{\psi} = \rho_0\left[\dot{e} - \dot{T}\eta - T\dot{\eta}\right]; \;\; \dot{\rho}_0 = 0, \tag{196}$$

where ρ_0 is the mass density at the reference configuration, such that $\rho_0 = J\rho$ for a system where mass is conserved. The transformation in Eq. (196) is intuitively explained, as the total internal energy is subtracted by a multiple of entropy by temperature. This transform is consistent to the adopted definition of entropy and it correctly captures the subtraction of dissipated energy off of the total energy to, finally, retrieve the free and available energy for usable work. The substitution of the Helmholtz free-energy density into the local form of the Clausius-Duhem inequality, Eq. (195), produces

$$-\rho \dot{T} \eta - \frac{1}{J} \dot{\psi} + \boldsymbol{\sigma} : \mathrm{grad}\,(\boldsymbol{v}) + \frac{1}{T} \mathrm{grad}\,(T) \cdot \boldsymbol{k} \mathrm{grad}\,(T) \geq 0. \tag{197}$$

Suppose that the Helmholtz free-energy density depends upon the left Cauchy-Green deformation tensor, $\boldsymbol{b} = \boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}$, the absolute temperature T, and a set of arbitrary tensors $\big\{\boldsymbol{z}_i \;\; \forall i \in \{1, 2, \ldots, n\}\big\}$ called *internal variables*. In mathematical format,

$$\psi = \psi\,(\boldsymbol{b}, T, \boldsymbol{z}_1, \boldsymbol{z}_2, \ldots, \boldsymbol{z}_n)\,. \tag{198}$$

The two first dependencies are well known to the reader at this point; the new concept, although, is that of internal variables. Dissipative processes, such as viscoelasticity, plasticity, and damage, are *history-dependent*—their current response at any time depends upon information that happened in its past. Thus, internal variables are tensor objects that store such information, as exemplified by plastic or viscous strain, damage size, etc.

Exercise 6.1

Given the left Cauchy-Green deformation tensor $\boldsymbol{b} = \boldsymbol{F}\boldsymbol{F}^{\mathrm{T}}$, show that

i) $\dot{\boldsymbol{b}} = 2\,\mathrm{sym}\big(\mathrm{grad}\,(\boldsymbol{v})\,\boldsymbol{b}\big)$;

ii) $\frac{\partial \psi}{\partial \boldsymbol{b}} \in \mathrm{Sym}^3$, given that $\psi = \psi\,(\boldsymbol{b})$.

Then, using Eq. (32), show that

$$\frac{\mathrm{d}}{\mathrm{d}t} \psi\,(\boldsymbol{b}) = 2 \frac{\partial \psi}{\partial \boldsymbol{b}} \boldsymbol{b} : \mathrm{grad}\,(\boldsymbol{v})\,. \tag{199}$$

Plugging Eq. (198) into the time derivative of Eq. (197) results in

$$\underbrace{-\rho \eta \dot{T} - \frac{1}{J} \frac{\partial \psi}{\partial T} \dot{T}}_{I} + \underbrace{\boldsymbol{\sigma} : \mathrm{grad}\,(\boldsymbol{v}) - \frac{1}{J} \frac{\partial \psi}{\partial \boldsymbol{b}} : \dot{\boldsymbol{b}}}_{II} - \underbrace{\frac{1}{J} \frac{\partial \psi}{\partial \boldsymbol{z}_i} \odot \dot{\boldsymbol{z}}_i}_{III} + \underbrace{\frac{1}{T} \mathrm{grad}\,(T) \cdot \boldsymbol{k} \mathrm{grad}\,(T)}_{IV} \geq 0. \tag{200}$$

where $\odot$ is an appropriate contraction operation to contract all the indices of the time rate of the i-th internal variable with the corresponding last indices of the partial derivative of the Helmholtz free-energy potential with respect to that internal variable.

We can guarantee that the second law of thermodynamics is **strongly** satisfied by making each term equal to or greater than zero. For instance, the term I can be easily forced to be zero,

$$-\rho\eta\dot{T}-\frac{1}{J}\frac{\partial\psi}{\partial T}\dot{T}=-\left[\rho\eta+\frac{1}{J}\frac{\partial\psi}{\partial T}\right]\dot{T}=0\ \ \forall\dot{T}\implies\eta=-\frac{1}{\rho_0}\frac{\partial\psi}{\partial T};\tag{201}$$

which means that the entropy is the *conjugate* of temperature through the Helmholtz free-energy density. Next, the term II can also be forced to be zero, and applying the time derivative to the dependency postulated at Eq. (198) (see Eq. (199) for algebraic manipulation), one gets

$$\boldsymbol{\sigma}:\mathrm{grad}\left(\boldsymbol{v}\right)-\frac{1}{J}\frac{\partial\psi}{\partial\boldsymbol{b}}:\dot{\boldsymbol{b}}=\left[\boldsymbol{\sigma}-\frac{2}{J}\frac{\partial\psi}{\partial\boldsymbol{b}}\boldsymbol{b}\right]:\mathrm{grad}\left(\boldsymbol{v}\right)=0\ \ \forall\mathrm{grad}\left(\boldsymbol{v}\right)\implies\boldsymbol{\sigma}=\frac{2}{J}\frac{\partial\psi}{\partial\boldsymbol{b}}\boldsymbol{b}.\tag{202}$$

Equation (202) possesses too large an importance to discuss it here and now. Nevertheless, one should observe that, for the very first time, we established a relation between stress and strain, i.e. a constitutive relation for stress. This will be discussed in the next section, since it will enable the construction of BVPs in terms of, at least, the displacement field.

Term IV, on the other hand, can be forced to be greater than zero, $IV > 0$, if the conduction tensor $\boldsymbol{k}$ is positive-definite. Get back to Eq. (58) and observe that IV is exactly a quadratic form as the one defined there. If one inspects the Duhamel conduction law, it becomes clear that positive-definiteness of the conduction tensor is not far-fetched. In fact, $\boldsymbol{k}$ must be a positive definite tensor to force the heat flux direction to remain in a cone opposite to the temperature gradient; which, in turn, maintains the natural direction of heat flux.

Finally, the only remaining constraint to force the strong satisfaction of the Clausius-Duhem inequality is

$$-\frac{1}{J}\frac{\partial\psi}{\partial\boldsymbol{z}_i}\odot\dot{\boldsymbol{z}}_i\geq 0\ \ \forall i\in\{1,2,\ldots,n\}\overset{J>0}{\Longrightarrow}\frac{\partial\psi}{\partial\boldsymbol{z}_i}\odot\dot{\boldsymbol{z}}_i\leq 0\ \ \forall i\in\{1,2,\ldots,n\}.\tag{203}$$

Although it seems that the four terms were arbitrarily chosen to be either null or greater than zero, it is not arbitrary at all. The easiest to explain is term IV, as the direction of heat flux must be inside a cone opposite to the temperature gradient to accord with the zero-th law of thermodynamics (flow of heat from the hotter to the colder regions).

The other terms underlined in Eq. (200) can be fully understood by imagining situations where temperature is constant in space and in time. For example, if there is no gradient or time rate of temperature, terms I and IV are automatically null; if the process is non-dissipative such as in elasticity, there is no sense in generation of net entropy nor is this process history-dependent, hence, term II must be unequivocally null. This procedure is known as the **Coleman-Noll procedure**. This thermodynamic procedure can be applied to a vast array of classes of constitutive models, both elastic and dissipative. For this reason, the Coleman-Noll procedure became standard, even for modern methods involving constitutive models based on machine learning, see [13, 14, 15, 16].

Exercise 6.2

i) Derive the second law of thermodynamics in material configuration.

ii) Derive the following constitutive relations for the first Piola-Kirchhoff stress tensor,

$$\boldsymbol{P} = \frac{\partial \psi}{\partial \boldsymbol{F}}, \tag{204}$$

and, for the second Piola-Kirchhoff stress tensor,

$$\boldsymbol{S} = 2\frac{\partial \psi}{\partial \boldsymbol{C}}. \tag{205}$$

6.2. Brief concluding commentary

This section showed one of the most mind-boggling fact of continuum mechanics to the student coming from a traditional background in mechanical engineering—thermodynamics and solid mechanics are not distinct disciplines; neither the study of separate phenomena. That assertion could not be further from the truth, since the determination of stress states is intimately linked to the thermodynamical state of the body.

The interplay of thermodynamics and continuum mechanics is fundamental in determining the physical consistency of constitutive models. We will see further into this text that there are more constraints for constitutive models to satisfy. However, we highlight that the second law of thermodynamics is such an important step that it is the core of more advanced constitutive modeling frameworks, such as variational constitutive models, see [17, 18] for instance.

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